

BCI STUDIO

Data Visualization Style Guide
Chart & table design standards

Version 1.0 | 2022



British Columbia Investment
Management Corporation



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Introduction

TRANSFORMING NUMBERS INTO STORIES

This is a practical guide for effective and consistent chart and table design at BCI.

You'll find formatting standards for charts and tables as well as best practices and strategies to clearly communicate your data story.

Elements like font and color, the widths of lines and bars, and the use of tick marks are all choices that determine whether a chart is clear, engaging, and consistent. This guide will help you make those choices.

Do you have a story behind your chart?

Data storytelling is the ability to effectively communicate insights from your dataset -- this guide will show you techniques on formatting your chart so that the key takeaway is the focus of your chart.

Who this guide is for?

This guide has been developed for the CIR Client Reporting team as a reference for preparing and setting content for the quarterly client reporting cycle.

The guidelines in this document are applicable to all charts and tables designed at BCI.

ACKNOWLEDGMENTS

There are many helpful references we've used to develop this guide:

Nancy Duarte *Data Story: Explain Data and Inspire Action Through Story*
www.duarte.com/datastory/

Ann K Emery, depictdatastudio.com

Stephanie Evergreen, stephanieevergreen.com/design-of-an-award-winning-report/

Eva Murray, Forbes.com; www.forbes.com/sites/evamurray/2019/04/04/why-do-bar-charts-work/?sh=31a382c0e43e

Cole Nussbaumer Knaflic *Storytelling with Data: a Data Visualization Guide for Business Professionals*.
www.storytellingwithdata.com/chart-guide

Steve Wexler *The Big Picture: How to Use Data Visualization to Make Better Decisions – Faster*.

John Schwabish, Policy Viz; <https://policyviz.com/2021/03/16/why-your-organization-needs-a-data-visualization-style-guide/>

Dona M. Wong *The Wall Street Journal Guide to Information Graphics: The Dos and Don'ts of Presenting Data, Facts, and Figures*.

Choose an effective visual, **keep it simple, avoid clutter and repetition, and highlight the key takeaway** in the chart



Colour palettes,
font styles & chart basics

Market Outlook

FIXED INCOME & CURRENCY

- 10-year bond yields in the U.S. are relatively flat. However, the yield on long-term bonds fell, resulting in the process of moving

Font styles & visual hierarchy

ORGANIZING CONTENT ON A PAGE

BCI uses the Calibri font family applied to the following font styles to create a visual hierarchy on a report’s page.

- Body text is left-aligned at BCI. Do not use full justification because it create gaps of whitespace between words, which can impact readability due to the uneven spacing and broken visual rhythm of the words.

Font style hierarchy adds structure to a page; **it creates visual organization, reinforces brand recognition**, and makes it easier for people to read and understand your content.

H1: Left-aligned, Calibri light 31pt, Midnight (#00365b), sentence case

INTRO TEXT: Left-aligned, Calibri regular 13pt, Midnight (#00365b), sentence case

H2: Left-aligned, Calibri bold 11pt, Dark ocean (#0091a1), uppercase

H3: Left-aligned, Calibri bold 11pt, Text black (#48484A), sentence case

BODY: Left-aligned, Calibri regular 11pt, Text black (#48484A), sentence case

SOURCE, NOTES, AND FOOTNOTES: Left-aligned, Calibri regular 8pt, Text black (#48484A), sentence case

H1 Heading/Headline

Vertical Spacing: 0.5”

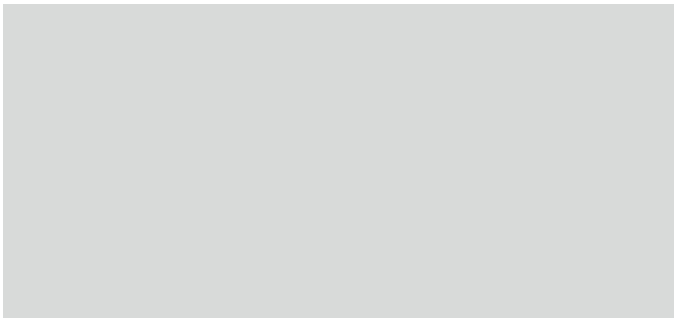
Intro text Lorem ipsum nem culliquo volum eturia seque eos nobiti beriate seque di occus idel ipsam ime

Vertical Spacing: 0.5”

H2 HEADING

H3 Heading

Body Text Lorem ipsum nem culliquo volum eturia seque eos nobiti beriate seque di occus idel ipsam ime nusaeresto od mi, soluptat eume eossimi, ut lam harum, volupta quiaeca borepel laborum quiatur.



Source: Acme Investments Corp.

Chart and table basics

Charts and tables help the audience understand complex data by presenting it clearly and neatly.

- Write your titles and sub-titles so that they emphasize the key message shown in the visual.
- Make font sizes for tables and chart labels, Calibri 9pt. Occasionally you may need to make smaller for more complex content. Overall, try to be consistent with font sizes.

GUIDELINES

Chart Width: a chart (or table) can be 1, 2, or 3 columns wide, following the column structure defined in the Wireframe Guide.

Figure #: (optional) identifies the chart or table being referenced in the narrative. Figure number should start at 1 on each page.

Title: Make it brief – 6 to 12 words preferably; it should express the idea or insight behind the chart.

Sub-title: (optional) Keep to two lines maximum; to expand on the chart’s main idea or key takeaway.

\$ and %: decrease the busyness of a chart or table by removing these symbols in their content, and instead add (\$B) or (%) after the chart title or table’s row heading.

Writing an informative title and sub-title will emphasize your chart or table’s key takeaway.

TITLE: Left-aligned, Calibri bold 11pt, Text Black, sentence case

SUB-TITLE: Left-aligned, Calibri 10pt, Text Black, sentence case

Y-AXIS TITLE: Calibri 9pt, Text Black, sentence case, horizontal

SOURCE, NOTES: Calibri regular 8pt, Text Black, sentence case

CHART WIDTH: 1, 2, or 3 columns wide in the BCI report template

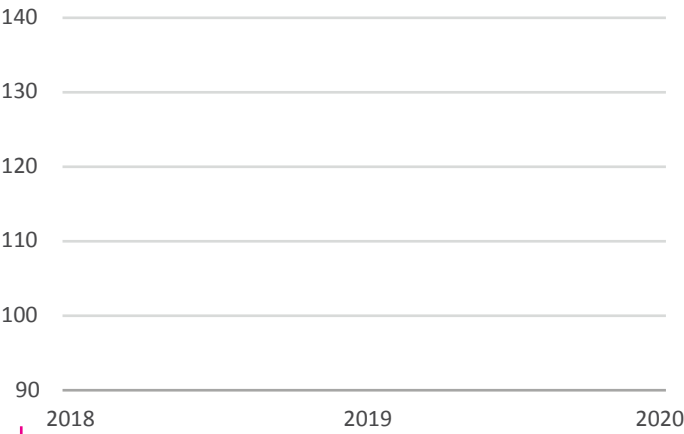
FIGURE #

Short catchy title with first word capitalized

Sub-titles can add explanatory power to a chart.

Vertical Spacing: 0.25"

Y-Axis title (units)



Vertical Spacing: 0.5"

Source: Acme Data Technologies
Notes: Vivamus nec iaculis ante, vel gravida lacus. Fusce in lectus non tellus interdum commodo. Curabitur et ligula in tortor facilisis auctor a vel magna.

Colours: chart data series

ORDER OF BCI BRAND COLOURS

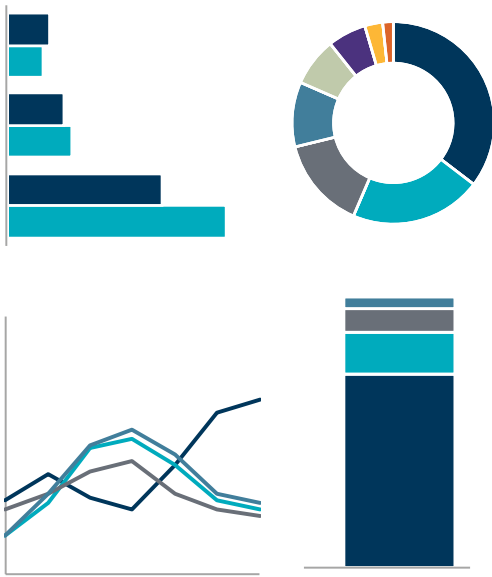
When depicting separate data series in a chart, use the colours in the order specified in the table to the left. Using colour consistently makes a report look organized and easier to follow.

- In bar, column, stacked, and donut charts, be sure to include a subtle separation between segments by adding a white outline. This creates a clearer delineation between categories.
- Best practice is to show four or fewer categories in a bar, line, or column chart; six or fewer in a donut chart. Plotting too many categories on the same chart makes comparing and contrasting difficult.
- Help your chart tell a story by pulling focus to one category over another. Use midnight (or ocean) on the feature segment; de-emphasize the other categories by applying Gray 3. See page 8 for more on this topic.

Using a **consistent order of colours** in chart design makes a report look organized and easier to follow.

BCI Colour Palette: Colours for data series bars, segments, & lines

BCI Colours		Hexcode	Series Order
	Midnight	#00365b	1
	Ocean	#00abbd	2
	Gray	#696f79	3
	Slate	#457b96	4
	Khaki	#a6b38c	5
	Purple	#4a317e	6
	Yellow	#fdb736	7
	Orange	#dc642b	8
	Emerald	#819fd4	9
	Jade	#57837b	10
	Amethyst	#864c9e	11
	Garnet	#ae1e57	12
	Melon	#f68e47	13
	Pear	#c1b64e	14
	Gray 3	#BFBFBF	De-emphasis



Colours: chart text & data labels

Adding data and text labels directly on the chart, rather than using a legend, will make your charts faster to read.

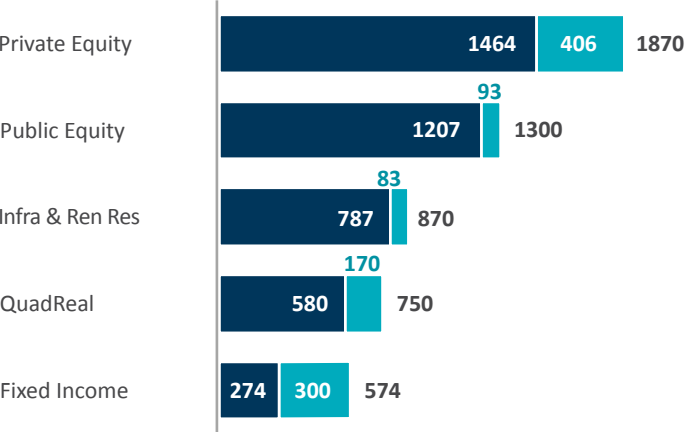
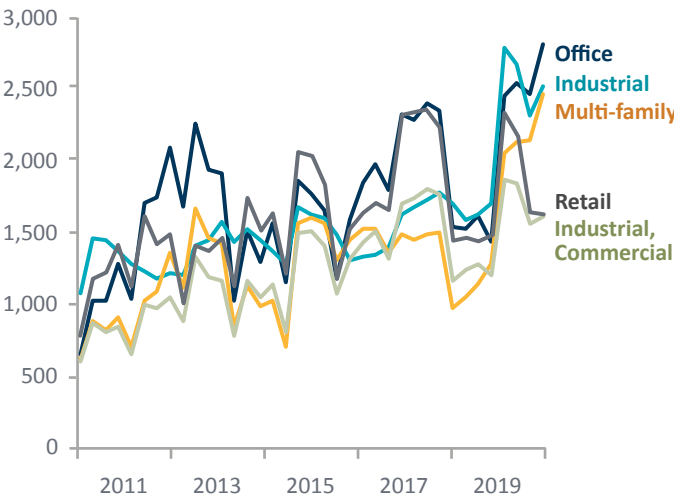
Where possible, make the label the same colour as the data series. However, note that **some of our colours require a darker version to ensure a strong contrast for readability and accessibility***. Those darker colours are for: ocean, khaki, yellow, and orange.

- Accessible design means **making sure that as many people as possible can enjoy and benefit from your work.**
- In some cases, it may be more appropriate to make the text or data labels in a chart **Text Black** or **white**, as in the example stacked bar chart to the right.

Applying **data and text labels in the colour of their series** will make charts faster and easier to read

Colours for data and text labels on a chart

BCI Colours		Hexcode
	Midnight	#00365b
	Dark Ocean*	#0091a1
	Gray	#696f79
	Slate	#457b96
	Dark khaki*	#808f57
	Purple	#4a317e
	Dark yellow*	#d07c18
	Dark orange*	#c74e23
	Text Black	#48484a



Colours: text and chart & table structures

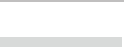
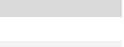
Referencing the colour table to the right, the darkest gray colour, Text Black, is the primary colour used for body text. This dark gray colour is easier on the eyes than pure black against a white background which can cause eye strain due to the extreme contrast from light to dark.

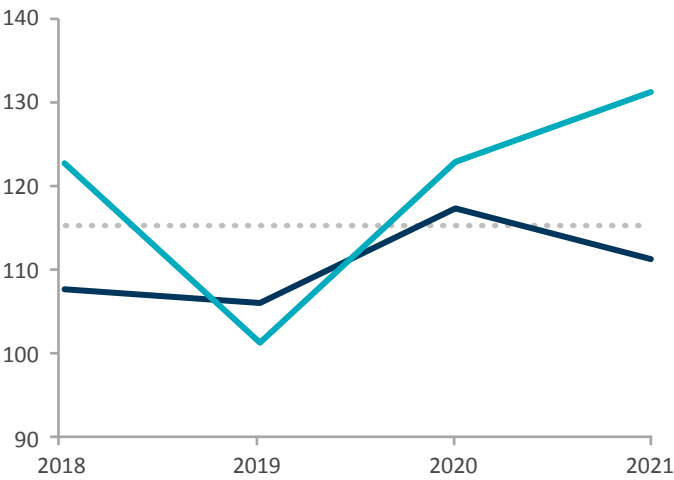
Use Text Black for overall body text in your report, and also for smaller text components in chart design, such as axis labels and titles, annotations, and source information.

- The lighter gray hues support the structures for charts and tables. Keeping the structure elements a lighter gray reduces the ‘noise’ of the chart.

Using **lighter gray tones on structures** will reduce the noise or busyness of a chart or table.

Colours for chart and table structures

BCI Colours	Hexcode		
	Text Black	#48484A	Text Black is used for report body text, table text, chart text and annotations, chart axis label text, chart axis titles, caption text, source text
	Gray 5	#7F7F7F	Data labels for de-emphasized data
	Gray 4	#A6A6A6	Chart x- and y-axis lines, tickmarks
	Gray 3	#BFBFBF	Horizontal table lines, de-emphasis for chart bar/line segments
	Gray 2	#D9D9D9	Chart gridlines, table row shading
	Gray 1	#F2F2F2	Chart shading, table column shading



ASSET CLASS	ASSET ALLOCATION EFFECT
TOTAL FUND	9.0
Leverage	-0.9
Infra. & Ren. Res.	0.2
Real Estate	4.0
Private Debt	-0.8
Mortgages	0.3
Bonds	6.5
Short-Term	-1.1

Storytelling with your data

WHEN YOU WANT YOUR CHART TO HAVE A CLEAR TAKE-AWAY MESSAGE

To pull focus to one category in a chart over another, use midnight (or ocean); and then de-emphasize the other categories by applying Gray 3.

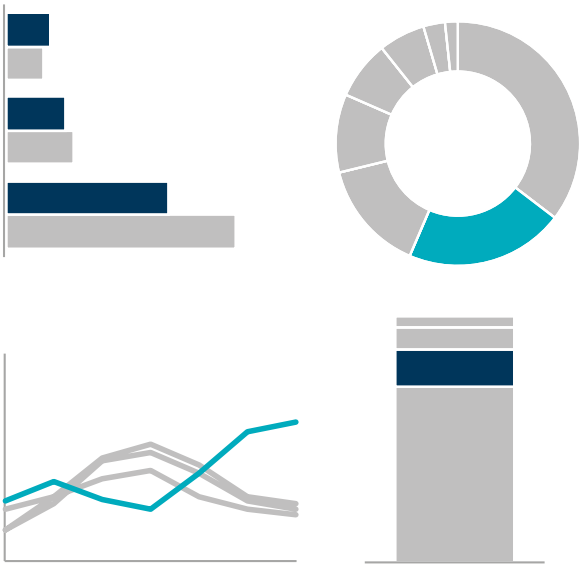
- In bar, column, stacked, and donut charts, be sure to include a subtle separation between segments by adding a white outline. This creates a clearer delineation between categories.
- For line charts, make sure the emphasized line is in front of all the other de-emphasized lines.
- Which colour should you use for emphasis – midnight or ocean? Midnight is the default BCI brand colour. However, both colours align with our brand – midnight is more formal, corporate, while ocean is more friendly and bright.

Think about what tone your chart should convey – that may help you decide which colour suits the message. Also consider consistency in your report, if you apply ocean in one chart, you may want to follow that design rule throughout the report.

BCI Colour Palette: Colours for data series bars, segments, & lines

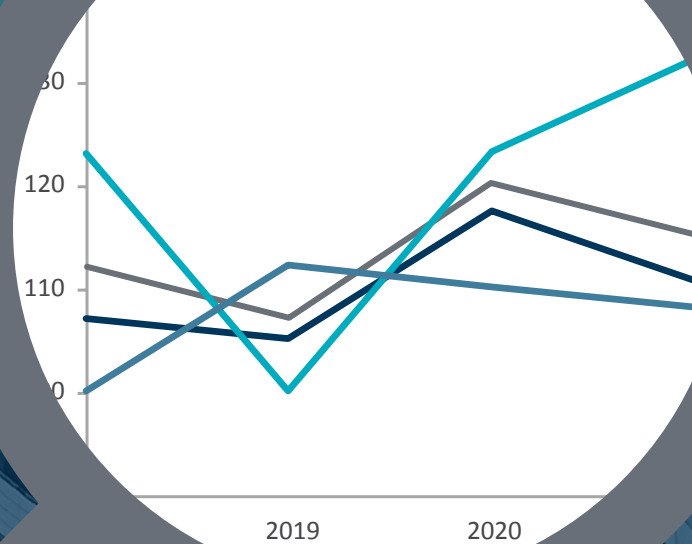
BCI Colours		Hexcode	Series Order
	Midnight	#00365b	Emphasis (option 1)
	Ocean	#00abbd	Emphasis (option 2)
	Gray 3	#BFBFBF	De-emphasis

Applying **colour strategically** will direct the focus to your chart’s main story



Line Charts

Tell a story by showing a trend or changes over time



Line charts

A line chart (time series chart) has an advantage over a bar or column chart because of its ability to show more information than just the final data point; a line chart shows historical activity. It can allude to the causation and history of the data, it can help end users quickly identify a trend, spot an outlier in a cyclical pattern, or analyze how a key metric is changing over time.

GUIDELINES

Gridlines: (optional) only use when precision in the interpretation of the values is needed; the approximate values that can be perceived without the aid of gridlines are almost always adequate. If not using a gridline, enable the Y-Axis Line.

Y-Axis: a line chart doesn't necessarily need a zero baseline. The purpose of a line chart is to show a trend, so choose a y-axis scale so that the height of the fever line occupies roughly two thirds of the chart area.

- **Y-Axis Title:** Keep brief, written horizontally, and left-aligned with chart.
- **Y-Axis Line:** (optional) if using horizontal gridlines, this is not needed.
- **Y-Axis labels:** Use increments people naturally use when counting: 0, 5, 10, 15; 0, 2, 4, 6, 8. Keep label values to four or less digits where possible.

X-Axis labels: keep it simple, avoid repetition (eg: Oct 2018, Nov 2018, etc.) instead, use two rows if necessary, months in first row, year in second row. Keep labels horizontal, do not place on an angle; use increments to avoid busy labels, such as 2016, 2018, 2020, 2022.

Tickmarks: x-axis labels can be placed below or between tickmarks; be consistent with your choice throughout the report

Fever line: Limit the number of lines to four or fewer. The purpose of a multiple-line chart is to compare

and contrast different data series. Plotting too many lines on the same chart gives a confusing picture and defeats the purpose. An alternative is to create an array of individual charts to allow better comparison. See example 7 on page 14.

Line colours: The order of colours to apply to the lines are noted on page 5.

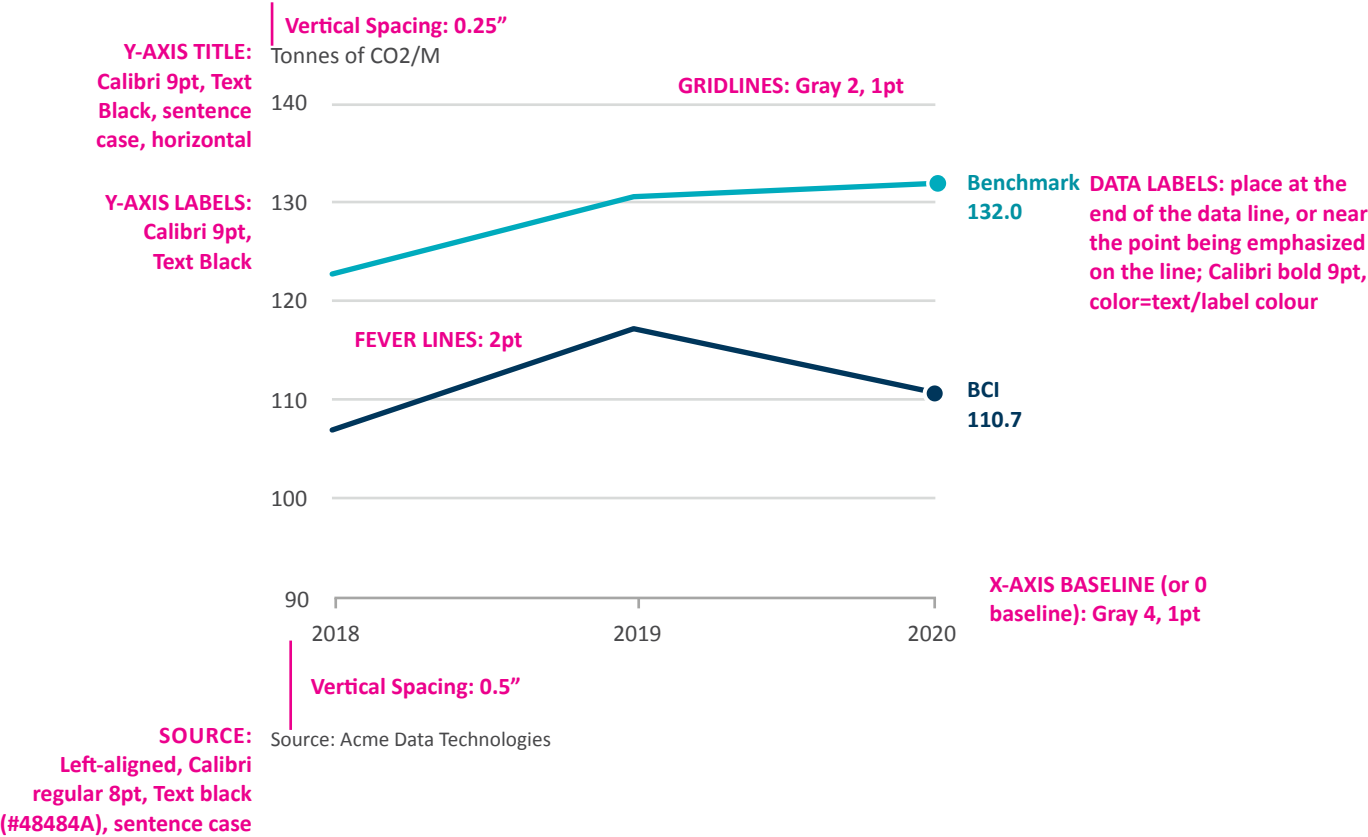
Data labels: label the lines directly, and in the darker colour version of the line. This allows the reader to identify the line and value quickly. See page 6 for label colours.

Legend: (optional) A legend is not required if the lines are labeled directly.

If a legend is required, place the legend right below the title area, using the respective text colours for each line; see example 4 on page 13.

FIGURE #
Short catchy title with first word capitalized

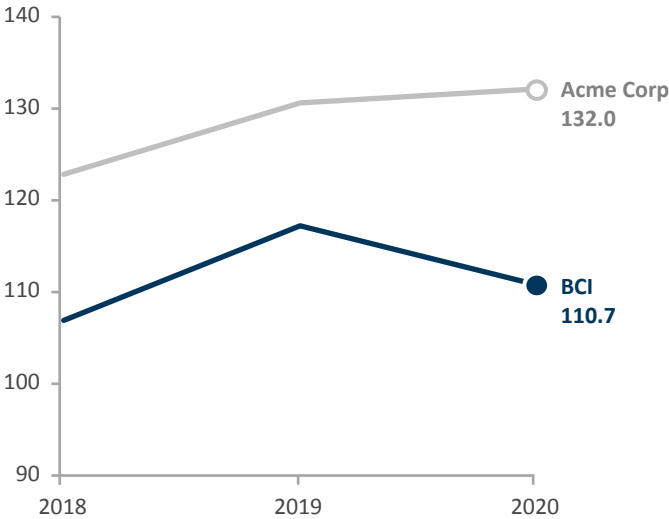
Sub-titles can add explanatory power to a chart.



Line charts are used to **display continuous data over a period of time**

Example 1

Focusing attention on one data series.



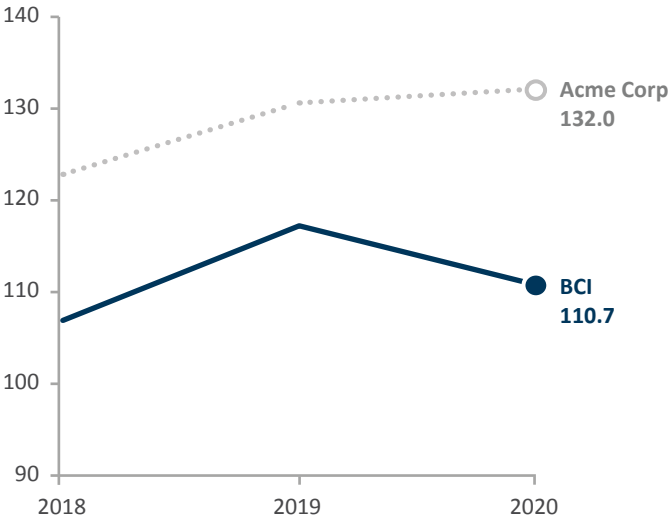
You can help your chart tell a story by focusing attention on the series that has the insight you’re presenting.

You can do that by applying the colour midnight to that series, making it stand out against the other(s). Colour the other lines Gray 3, their labels in Gray 5.

This chart has also been simplified by removing the grid lines. Grid lines aren’t necessary unless more precision is needed for discerning specific values.

Example 2

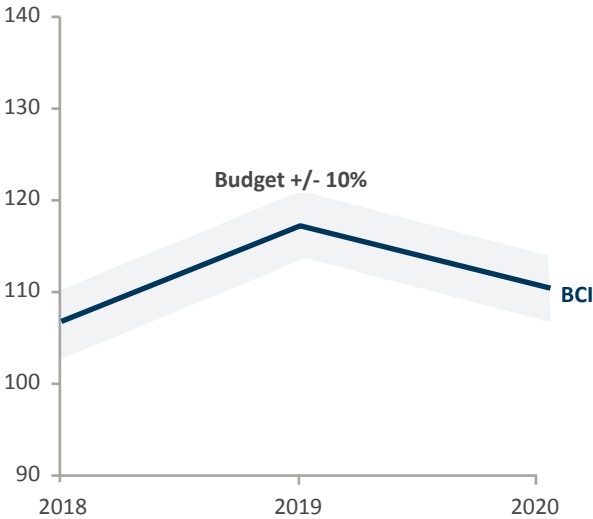
Comparing a data series against a target



Use a dotted line in Gray 3 and its label in Gray 5, to indicate a target, goal, or future state.

Example 3

Highlighting a range in a line chart

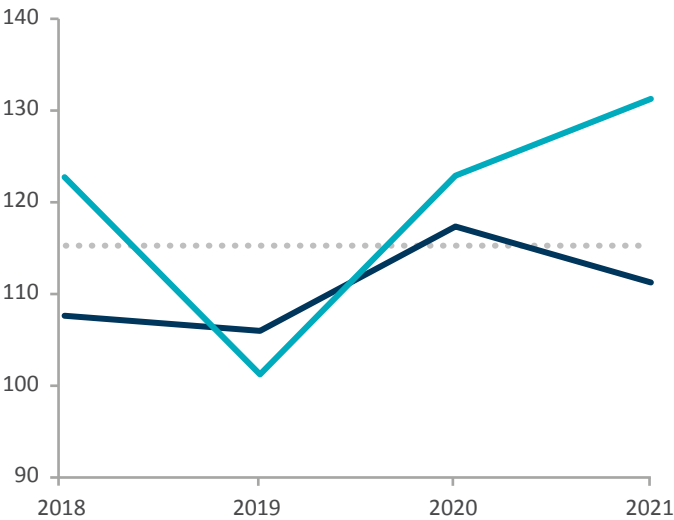


Apply a shading in Gray 1 to indicate a range.

Example 4

Using a legend rather than direct labeling

BCI | Acme Corp | Target

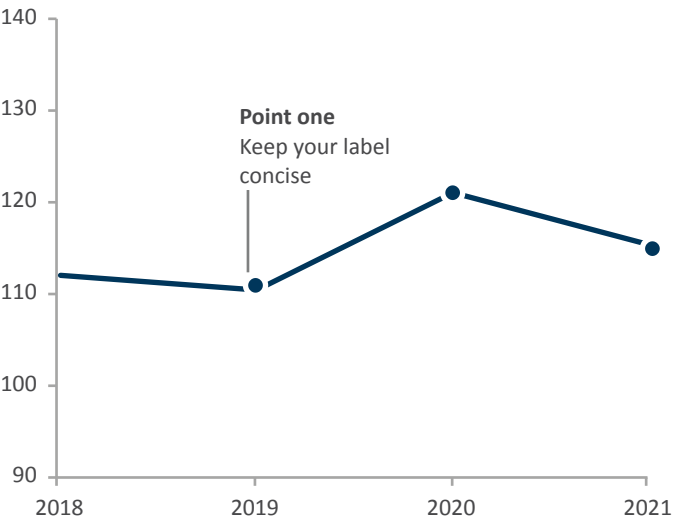


An alternative to directly labeling the data series, is to use a legend. Placing the legend right below the title area, using the respective text colours, is a simple quick reference for the viewer, font size 9pt.

Note the dotted line is Gray 3, but corresponding data label for “Target” is Gray 5. For some data series colours, text labels need to be darker than their segment colours for legibility. See page 6 for more about text label colours.

Example 5

Annotating on the data series

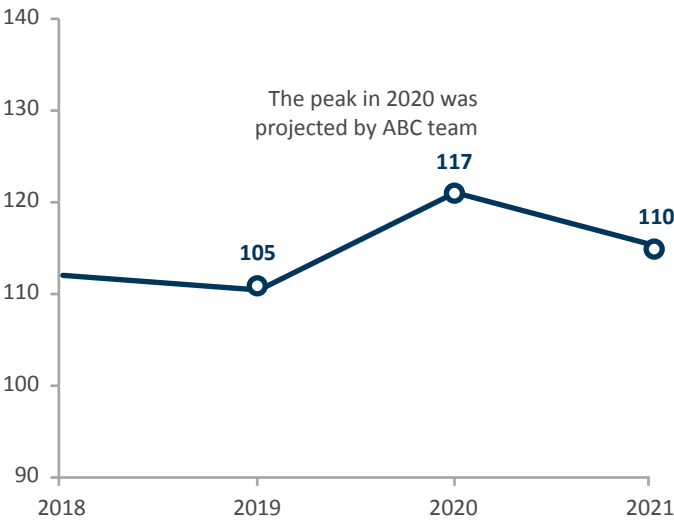


Annotations must be concise, no more than one short sentence written in text black. A large block of text will overwhelm the line. The font size should be 9pt (or 8pt if necessary), consistent with the chart’s other text labels.

Only use markers when featuring a point (or a few points) on a chart.

Example 6

Directly labeling data values



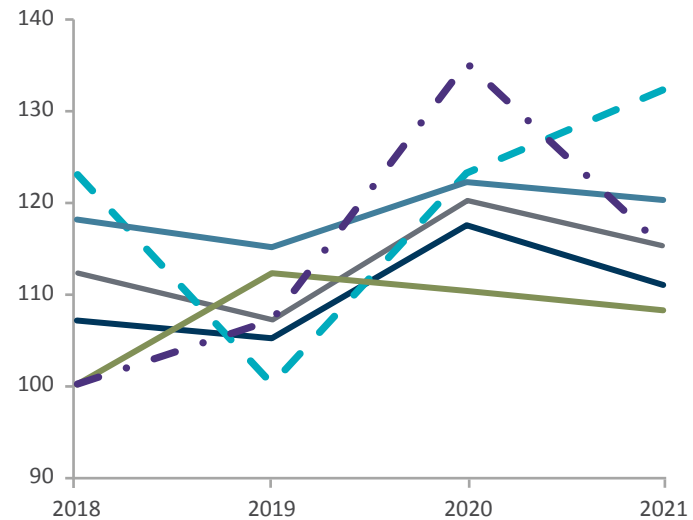
Line charts are suited for showing trends and volatility, and don’t typically need each point labeled. Keep labels consistently formatted and positioned near their markers.

Markers are white with an outline of 2pts in the line colour – or vice versa – line colour with white outline. Either option falls within BCI’s brand, however, choose a preference and be consistent throughout the report.

Example 7

Avoid plotting more than four lines on a chart; instead create an array of individual spark-line style charts.

BEFORE



Try to avoid showing more than four lines of data on a chart: keep the maximum number of lines to three or possibly four if the lines aren't intersecting at too many points. Plotting too many lines on the same chart gives a confusing picture.

An alternative to the above busy chart, is to create an array of individual charts to allow better comparison. The line colours can all be the same colour, the y-axis can be simplified.

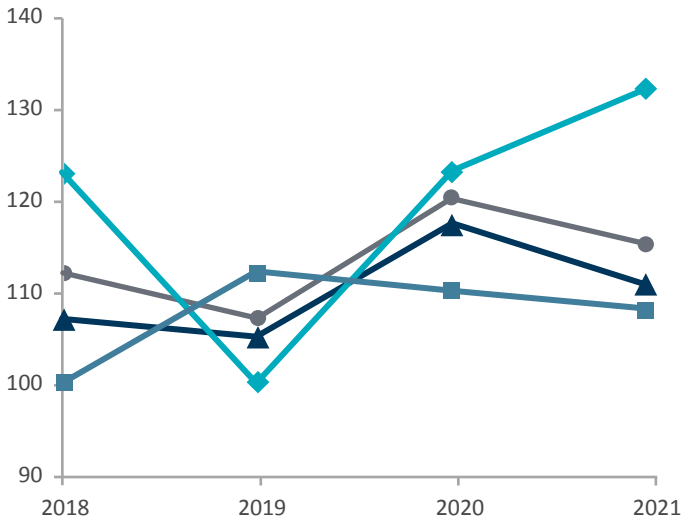
AFTER



Example 8

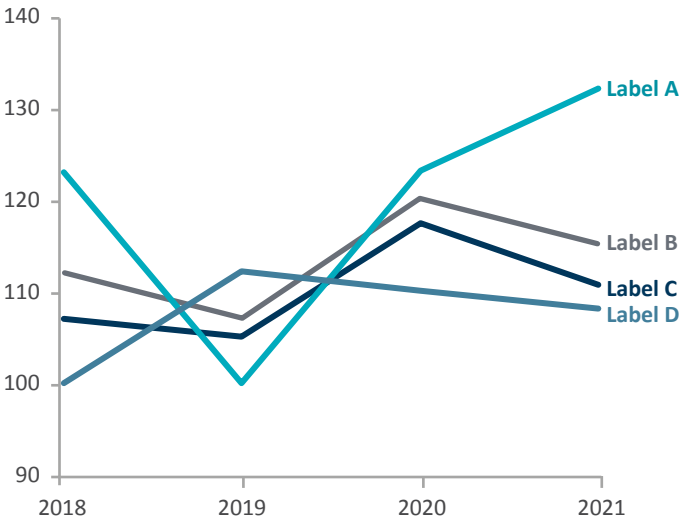
Avoid using data markers on every point and line.

BEFORE



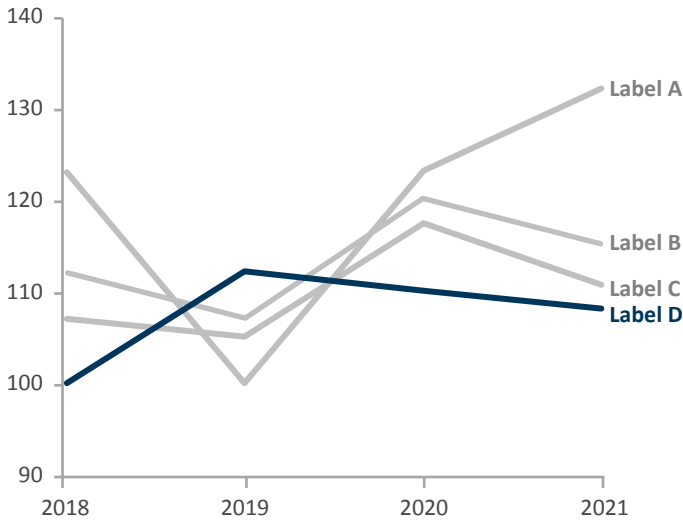
It is tempting to try out various line shapes, colours, and markers to make each line stand out. However, the overall effect can make your chart busy and a challenge to read.

AFTER (OPTION 1)



Instead use direct labeling, writing the label in the respective text colour of each line.

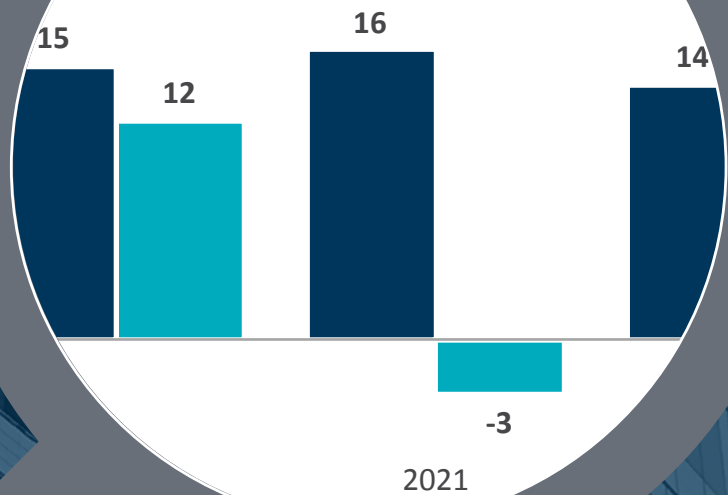
AFTER (OPTION 2)



An even simpler approach, is to pull focus to one of the lines, make the others Gray 3, while still using direct labeling in Gray 5. This approach is a great way to emphasis your chart's story.

Column Charts

Tell a story by comparing & contrasting heights



Column Charts

GUIDELINES

Vertical columns are great for comparing and contrasting values based on height.

Grid lines & Y-axis: neither are required if data labels are placed at the end of each bar/column.

Zero baseline: It's important that column charts always have a 0 baseline, otherwise the chart is misleading – the truncation hides the total value of each bar, making comparisons difficult.

Columns: The width of the columns should be about twice the width of the space between them. If you have more than one category, add 2pts of whitespace between columns within a category.

Data Labels: at the end of each column – can be in Text Black or in the series' respective text colour.

Legend: Put the legend at the top of the plot area for quick reference; text label colours should correspond with their respective bar colours. See page 6, for the text label colour guide.

- Column charts can become difficult to read if there are too many categories; try to limit to four.

- If the x-axis series labels are very long and don't fit horizontally, do not position at an angle. Instead use a bar chart to depict your data. Bar charts offer ample room for series labels.
- While you may have chosen a column chart to show a value at a particular point in time, consider instead a line chart – also known as a time series chart. A line chart can show activity over time, and is the ideal tool for helping the audience quickly identify a trend, spot an outlier in a cyclical pattern, or analyze how a key metric is changing over time.

Column charts show **comparative data clearly**, and are easy for your audience to understand 'how many' or 'how much'.

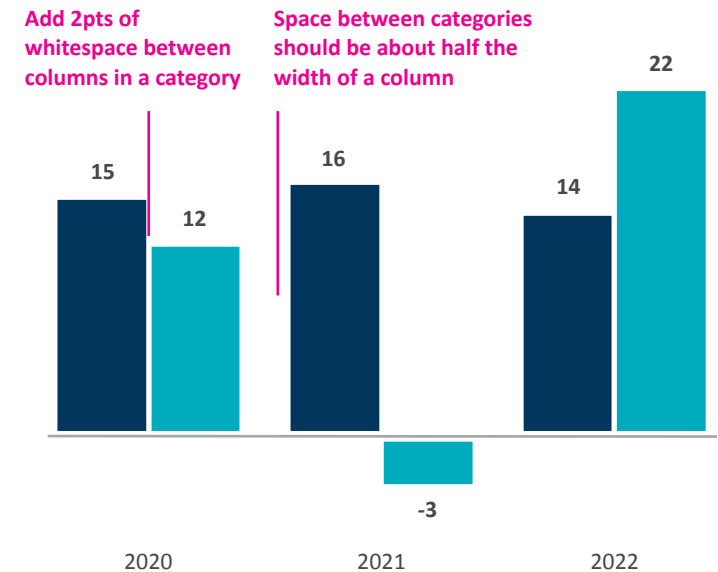
FIGURE #

Short catchy title with first word capitalized

Sub-titles can add explanatory power to a chart.

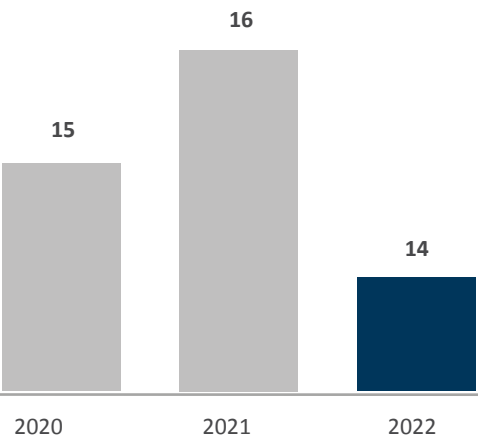
Victoria Office | **Vancouver Office**

LEGEND: Calibri 9pt, apply text colours same as category colours (see page 6)



Example 1

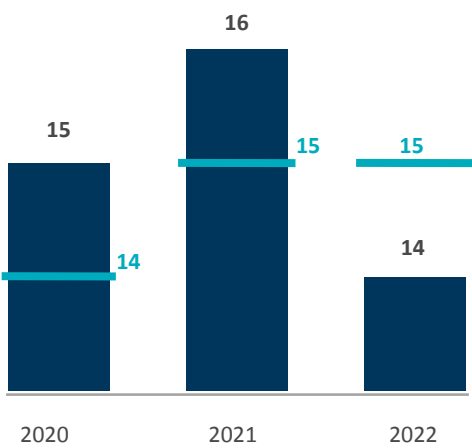
Focusing attention on one data series.



Use the colour midnight to make one column standout against the other(s). Colour the other columns Gray 3.

Example 2

Comparing results against a benchmark

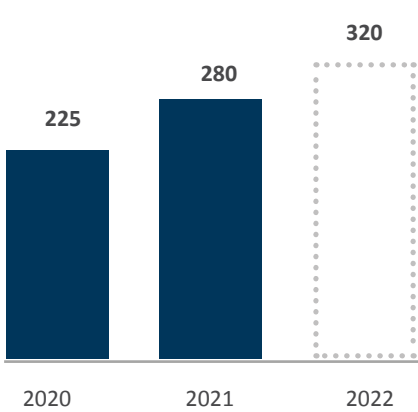


Increase the gap between columns from 50% column width to 75%+, to allow for benchmark labels.

Benchmark lines on column charts are coloured ocean, and their labels are in dark ocean. See page 6 for text label colours. Benchmark lines should be 2.5-3pts thick.

Example 3

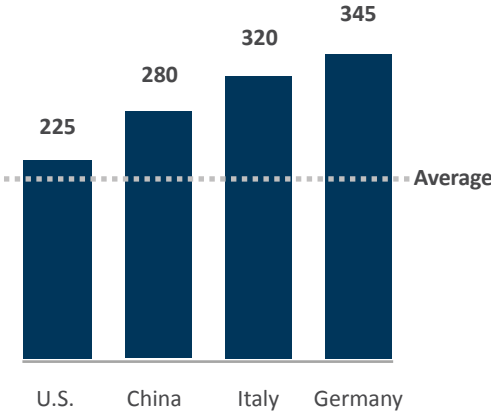
Showing an estimate using a dotted line, Gray 3, 2pts



Use a dotted line to indicate an estimate or projection

Example 4

Show data in ranking order for simplest visual comparison

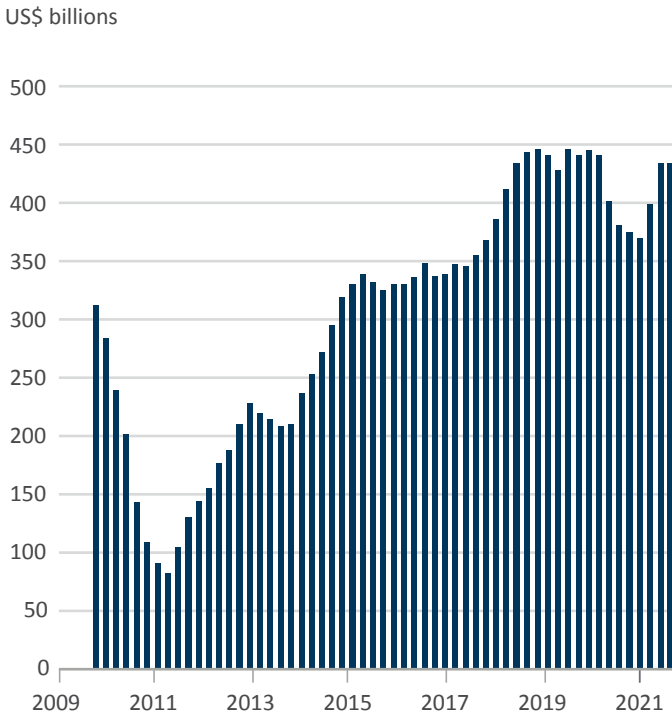


If the data being presented isn't in chronological order, present the bars in ranking order: while it is always possible to compare the bar lengths no matter the order, this can reduce the burden on the reader to make those comparisons themselves.

Example 5

Alternatives to displaying many columns in a narrow space

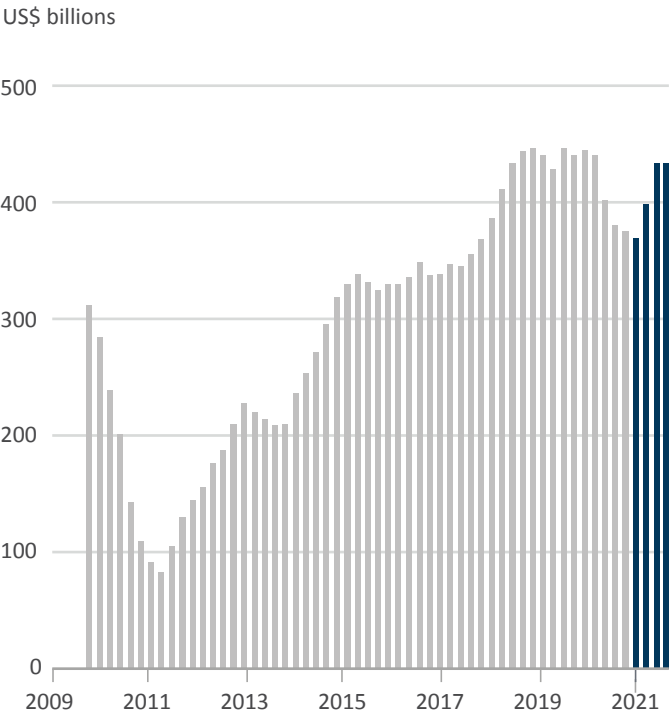
BEFORE



Depicting many columns in a narrow space can sometimes appear overly busy, or make that part of the page more prominent than other information on the page.

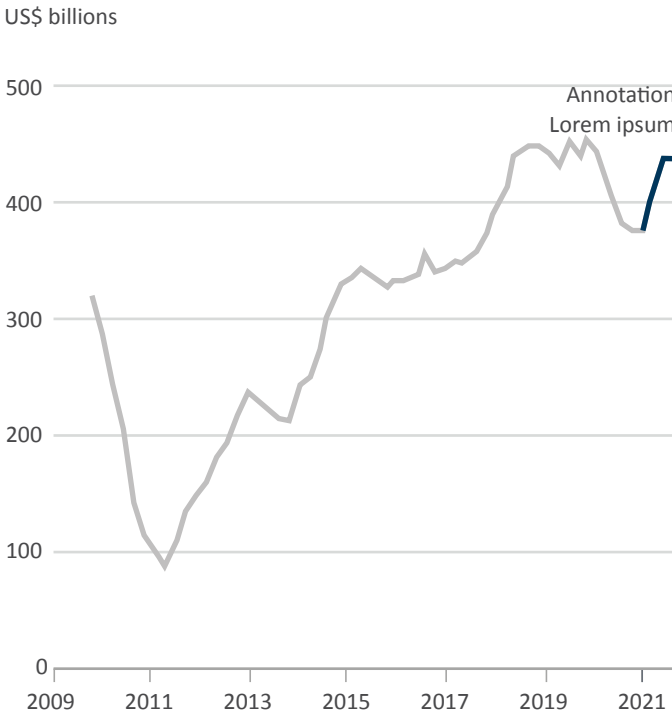
This could be an opportunity to adjust the formatting to highlight the key takeaway in the data.

AFTER (OPTION 1)



Applying the colour midnight emphasizes the primary data columns, in contrast to the others, which are de-emphasized with Gray 3.

AFTER (OPTION 2)



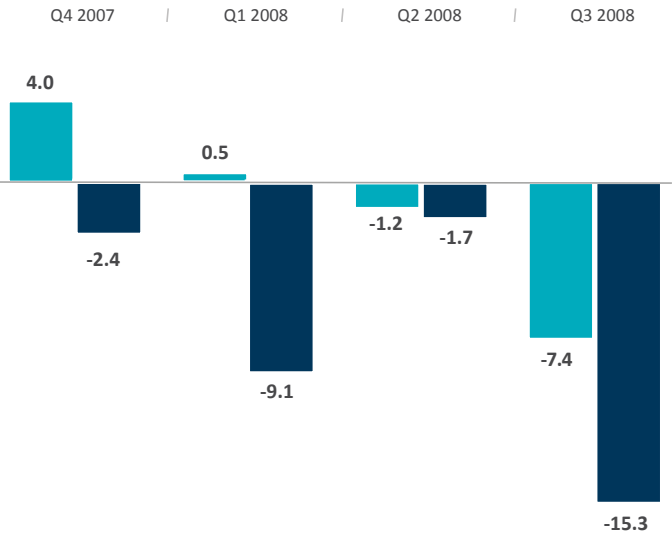
A line chart replaces the column chart, and the line colour changed to midnight on the segments being featured.

And there is room for an annotation about the key takeaway in the plot area

Example 6

Putting x-axis labels at the top of a column chart

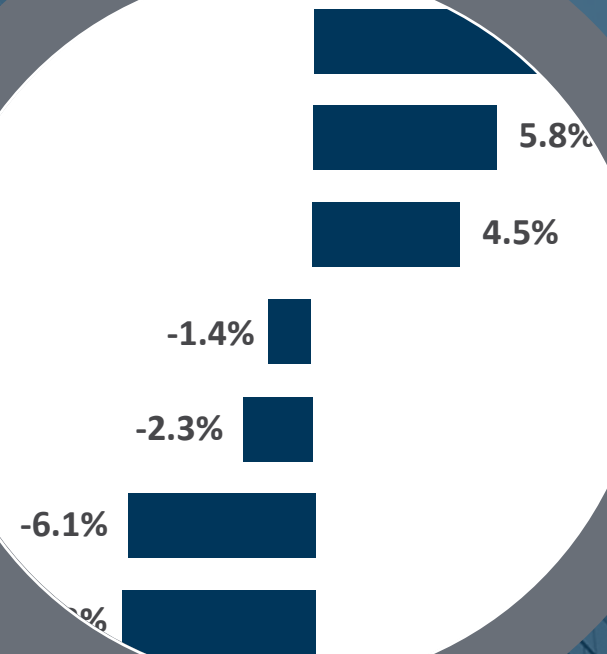
MSCI World Index | Private Equity



Put the x-axis labels at the top of a column chart when most of the values in the column chart are negative. The labels are nearer to the categories, so will be easier to associate with their corresponding values.

Bar Charts

Tell a story by comparing & contrasting lengths



Bar Charts

Bar charts are better for representing data with long labels as opposed to column charts.

GUIDELINES

Bars: The width of the bars should be about twice the width of the space between them. If you have more than one category, add 2pts of whitespace between bars within a category.

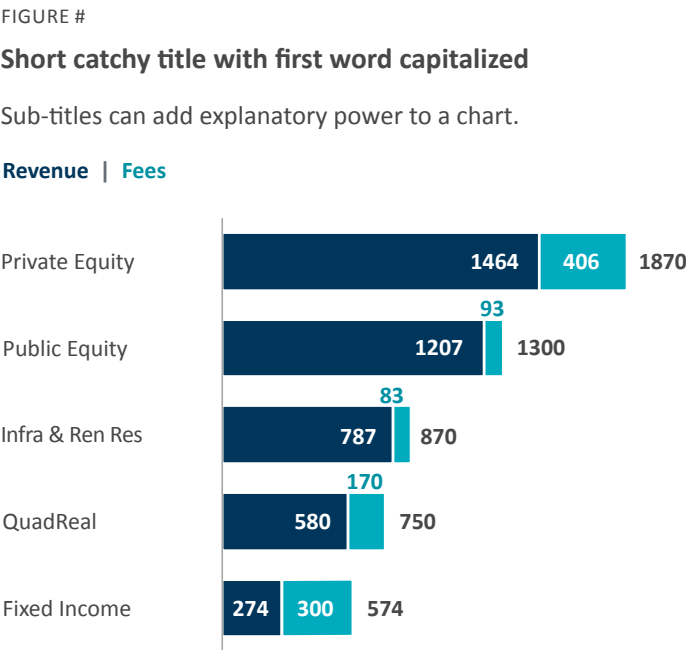
Value Labels: at the end of each bar – can be in Text Black or in the series’ respective text colour.

Bar labels: left-alignment is preferred but not required; this alignment is consistent with the overall columns of text alignment on a page – and keeps a page looking cleaner.

Legend: If plotting more than one category, put the legend at the top of the plot area; text labels should correspond with their respective bar colours. See page 6, for the text label colour guide.

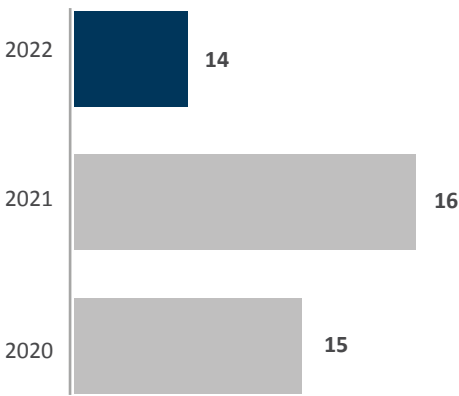
- Bar charts can become difficult to read if there are too many categories; limit to four.
- If the data isn’t chronological, sort the data in ranking order (ascending or descending) for ease of length comparison.
- It’s important that bar charts always have a 0 baseline, otherwise the chart is misleading – the truncation hides the total value of each bar, making comparisons difficult.
- While you may have chosen a bar chart to show a value at a particular point in time, consider instead a line chart – also known as a time series chart. A line chart can show activity over time, and is the ideal tool for helping the audience quickly identify a trend, spot an outlier in a cyclical pattern, or analyze how a key metric is changing over time.

Bar charts show comparative data clearly, while also providing more space for long labels



Example 1

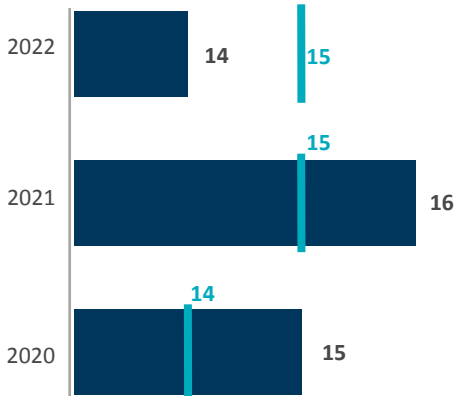
Focusing attention on one data series.



Use the colour midnight (or ocean) to highlight a category. Colour the other segments Gray 3. See page 8 for more information about focusing attention on a data series.

Example 2

Comparing results against a benchmark

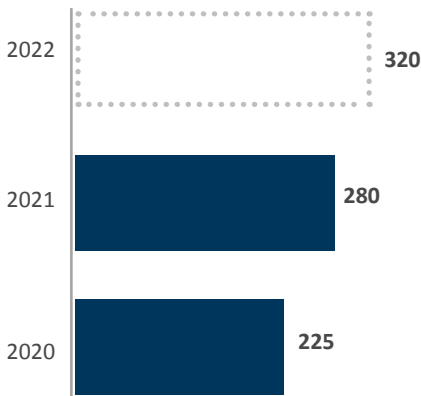


Increase the gap between bars from 50% column width to 75%, to allow for benchmark labels.

Benchmark lines on bar charts are coloured ocean and are 2.5-3pts thick; their labels are in dark ocean. See page 6 for text label colours.

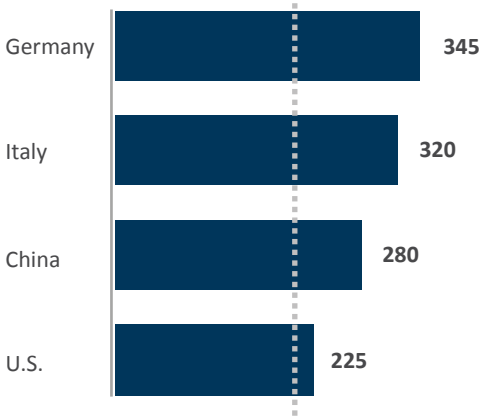
Example 3

Showing an estimate or projection by using a dotted line, Gray 3, 2pts



Example 4

Show data in ranking order for simplest visual comparison

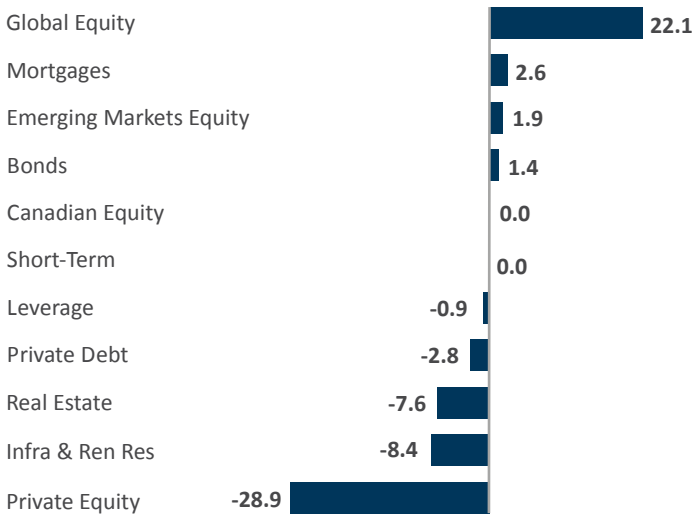


If the data being presented isn't in chronological order, consider presenting the bars in ranking order: while it is always possible to compare the bar lengths no matter the order, this can reduce the burden on the reader to make those comparisons themselves.

Example 5

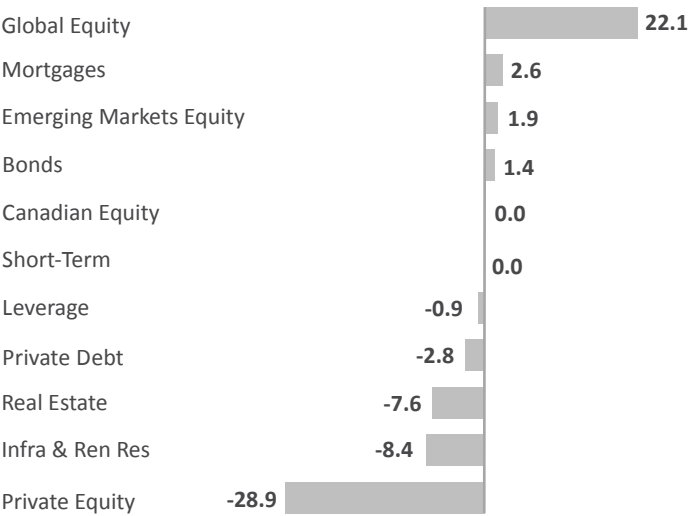
Alternatives to displaying many bars

BEFORE



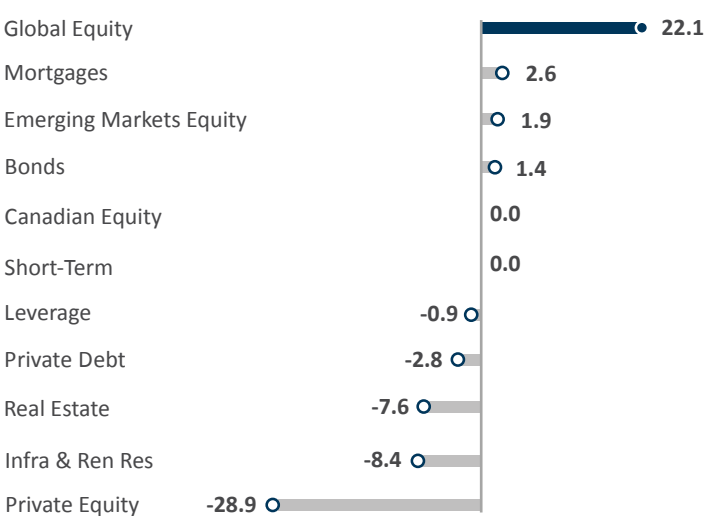
Depicting many bars in a narrow space can sometimes appear overly busy, or make that part of the page more prominent than other information on the page.

AFTER (OPTION 1)



Consider instead option 1, drawing focus to the segment of most interest in the chart; or option 2, de-emphasizing the ‘visual weight’ by narrowing the bars colouring them gray 3, and adding a circle at the end to emphasize the end point.

AFTER (OPTION 2)

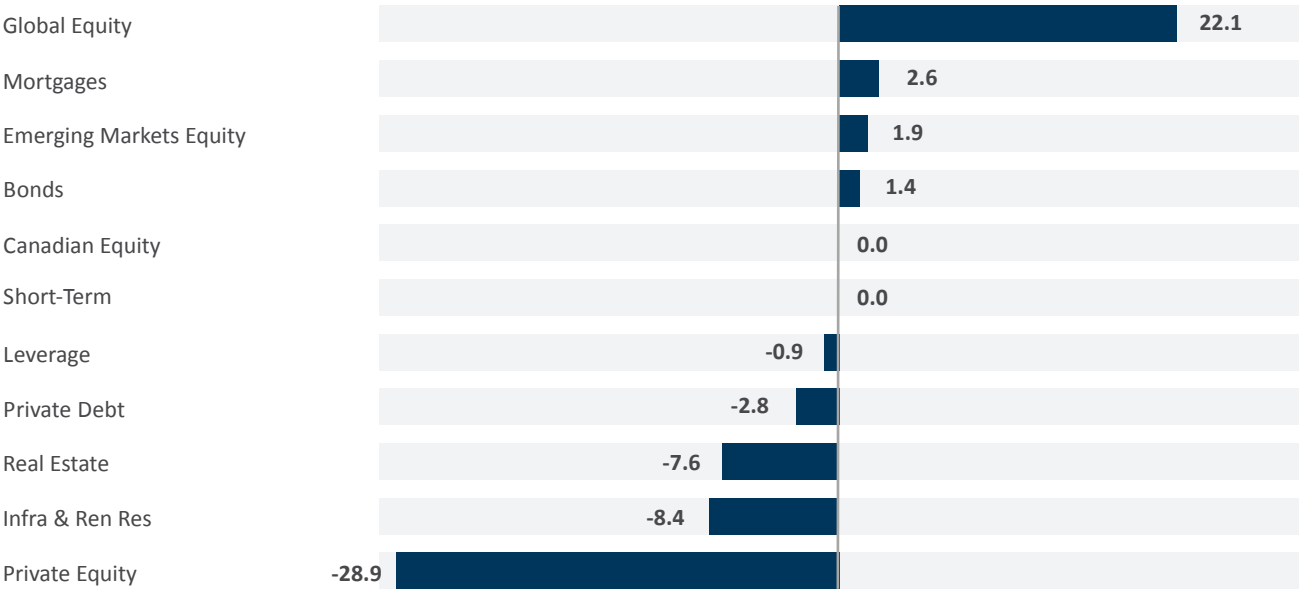


Emphasis is placed at the end of each bar, by adding a blue circle; this is known as a lollipop chart.

Example 5

AFTER (OPTION 3)

Adding light gray bars to create subtle structure for readability

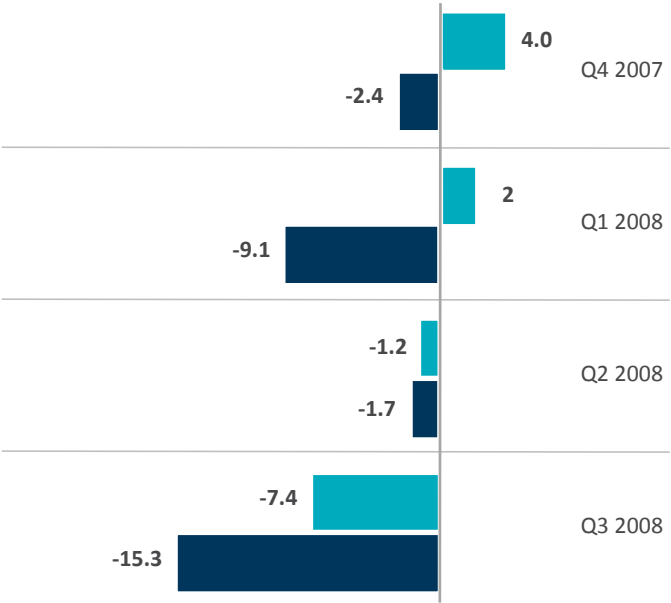


The subtle gray bars (Gray 1), help guide the reader’s eye and creates an overall clean structure for the bar chart

Example 6

Putting the labels on the right side of a bar chart

MSCI World Index | Private Equity

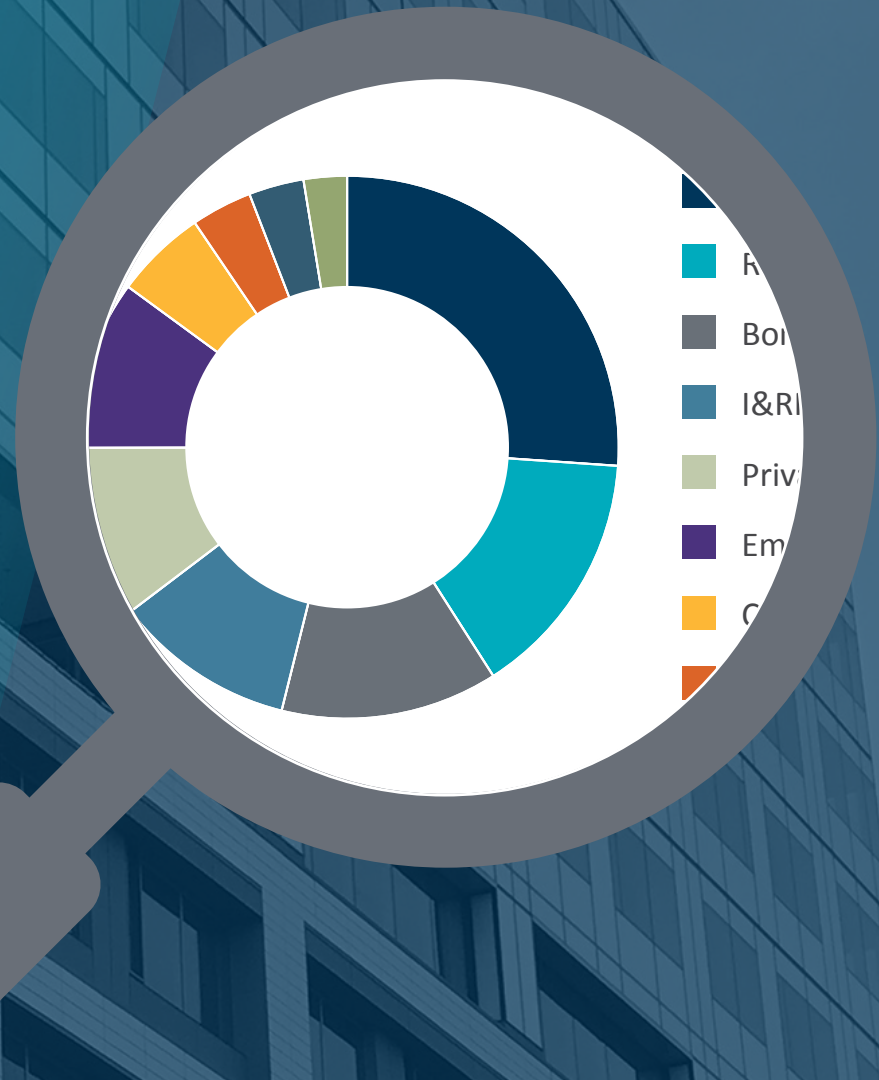


Put the x-axis labels along the right side of a bar chart when most of the values in the bar chart are negative. This way the labels are nearer to the categories, so will be easier to associate with their corresponding values.

When you have more than one category, it may help to separate with a thin line in Gray 3.

Donut Charts

Tell a story by comparing parts to a whole



Donut Charts

While donut charts help to show a part-to-whole relationship, they can become crowded and hard to read if there are too many segments.

- Donut charts are most effective when depicting 6 categories or less.
- Don't extract a segment from the donut chart, or display the chart in 3D. Comparing segment sizes is a challenge at the best of times.

GUIDELINES

Segment order: order segments from largest to smallest, clockwise from top centre position.

Segment colours: The order of colours to apply to the segments are noted on page 5.

Separate segments with a thin white outline – to clearly distinguish each segment.

Donut hole: is about 65% diameter.

Legend or Direct labels?

Legend (recommended approach) The example top right shows the legend styling:

- colour swatches are large enough to be clearly indicative of colour of the segment;

- values are right-aligned
- labels positioned 3 spaces after value – note the clean alignment of all three columns in the legend.

The advantage of a legend is the clean layout of the information, and overall consistent look when you have more than one donut chart in your report. The disadvantage is the extra step for the viewer to associate the swatch with the segment.

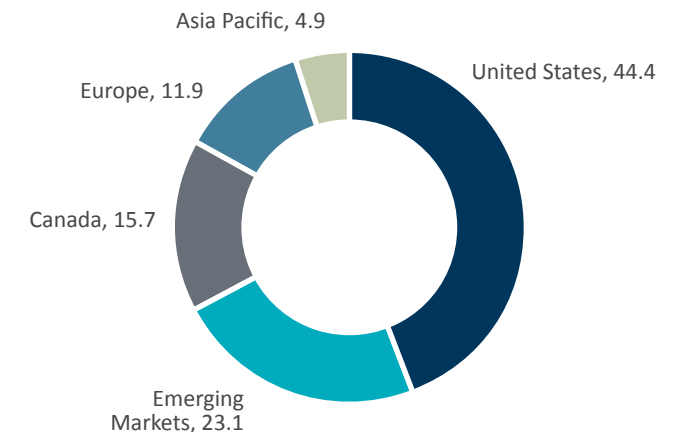
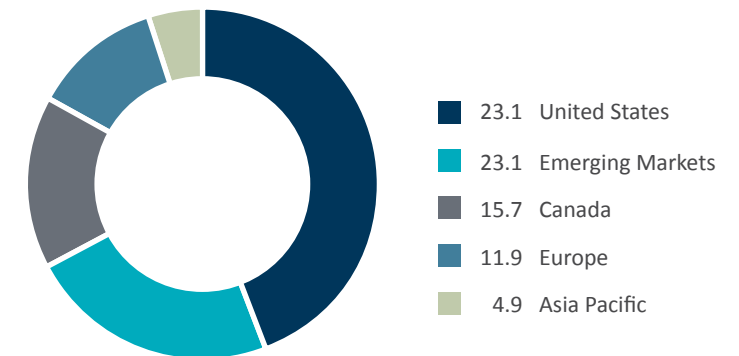
Direct Labels Labels are positioned close to their segments (bottom right), with consistent spacing between the label and the segment.

The advantage of this layout is clear association of label and value with segment.

The challenge can be a busy looking chart; and, maintaining a consistent look for all donut charts in the report. The more segments, the busier the chart, especially in the top half of the chart, with the smaller segments.

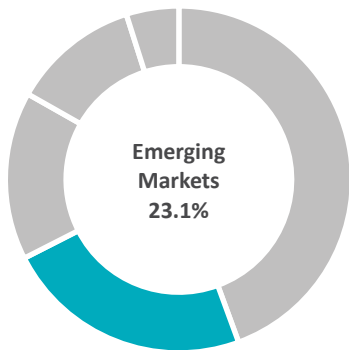
Directly labeling segments means that donut charts may take up a different amount of space from one donut to the next in your report – which can look busy or distracting.

Add 1-2pts of whitespace between segments for clarity



Example 1a

Focusing attention on one category



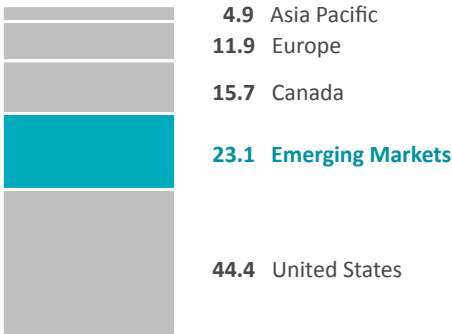
Use the colour midnight (or ocean) to highlight the important segment. Colour the other segments Gray 3. See page 8 for more information about focusing attention on a data series.

The highlighted segment doesn't have to be the largest slice. However, do not reorder the segments.

If you like, you can display the segment label in the centre of the donut.

Example 1b

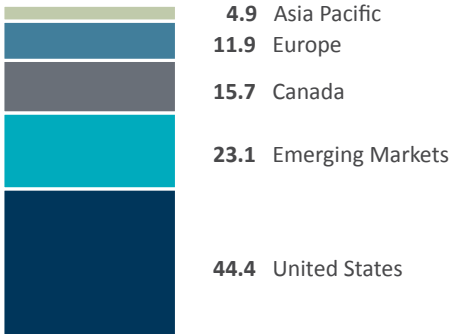
Focusing on one category



The highlighted segment is coloured ocean, and the secondary segments in Gray 3. Arrange the segments from largest to smallest, starting from the bottom up.

Example 2a

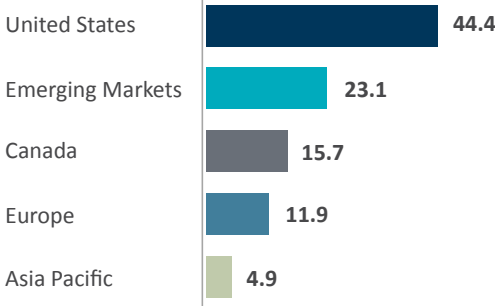
An alternative to the donut chart, is a 100% Stacked Chart, rank the data from bottom baseline, to top



Segment sizes are easier to compare in a 100% stacked chart, than in a donut or pie chart.

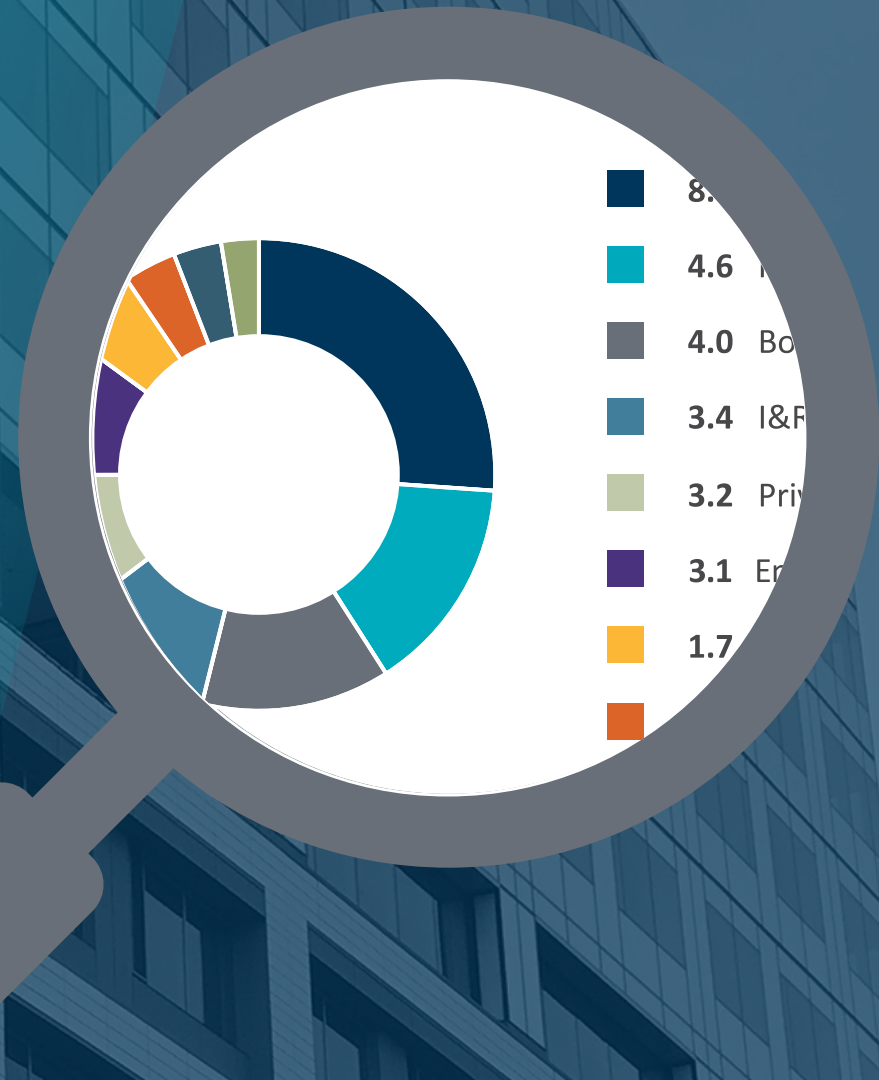
Example 2b

An alternative to the donut chart, bar chart, ranking data in ascending or descending order



Stacked Charts

Tell a story by comparing parts to a whole



Stacked Charts

GUIDELINES

Stacked charts show the relationship of each part to its whole.

Bars: The width of the bars should be about twice the width of the space between them.

If needed, increase the gap between bars from 50% column width to 75%+, to allow for labels as in the example to the right, where some bars are too narrow to hold the label.

If you have more than one category, add 2pts of whitespace between columns/bars within a category.

Bar labels: left-alignment is preferred but not required; this alignment is consistent with the overall columns of text alignment on a page – and keeps a page looking cleaner.

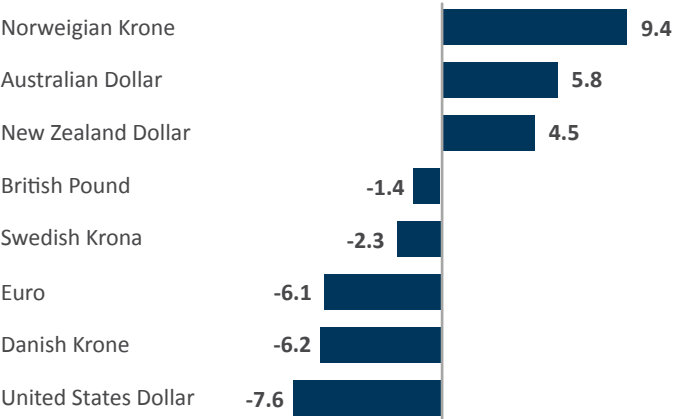
Data Labels: inside, at the end of each bar – in white, and if adding a grand total at the outside end, colour in text black.

Legend: Put the legend a the top of the plot area; text labels should correspond with their respective bar colours. See page 6, for the text label colour guide.

- If the data being presented isn't in chronological order, consider presenting the bars in ranking order, for ease of comparison.
- Stacked charts can become difficult to read if there are more than four sub-bars within a main bar.
- Consider which of the categories will be the primary and have it adjacent to the baseline – for ease of comparison.
- It's important that the chart always has a 0 baseline.

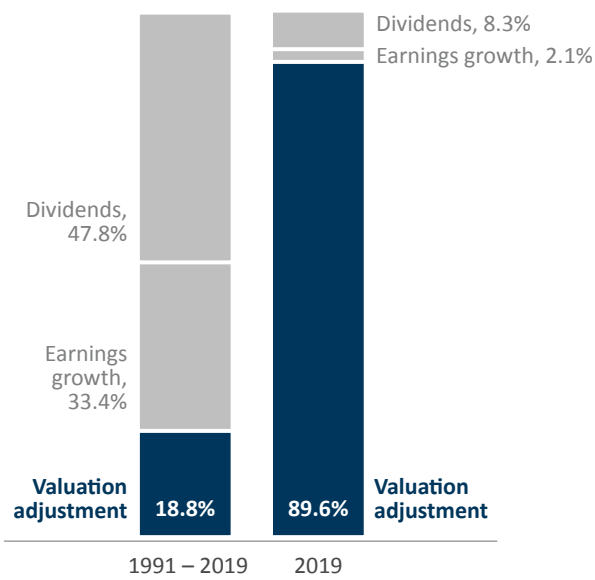
FIGURE #
Year-to-date performance (%)

Sub-titles can add explanatory power to a chart.



Example 1

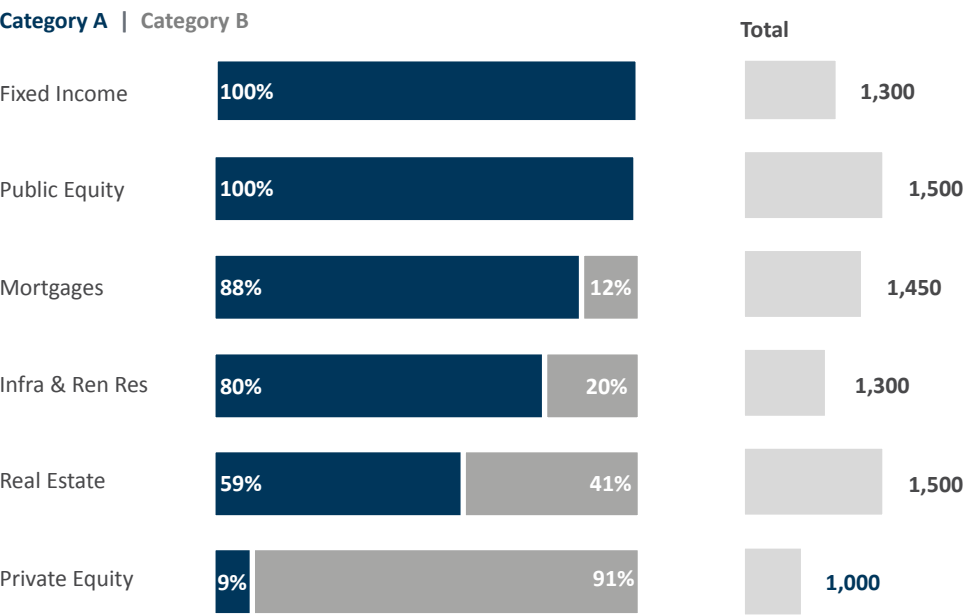
Focusing attention on one data series.



Use the colour midnight (or ocean) to highlight the important segment. Colour the other segments Gray 3. See page 8 for more information about focusing attention on a data series.

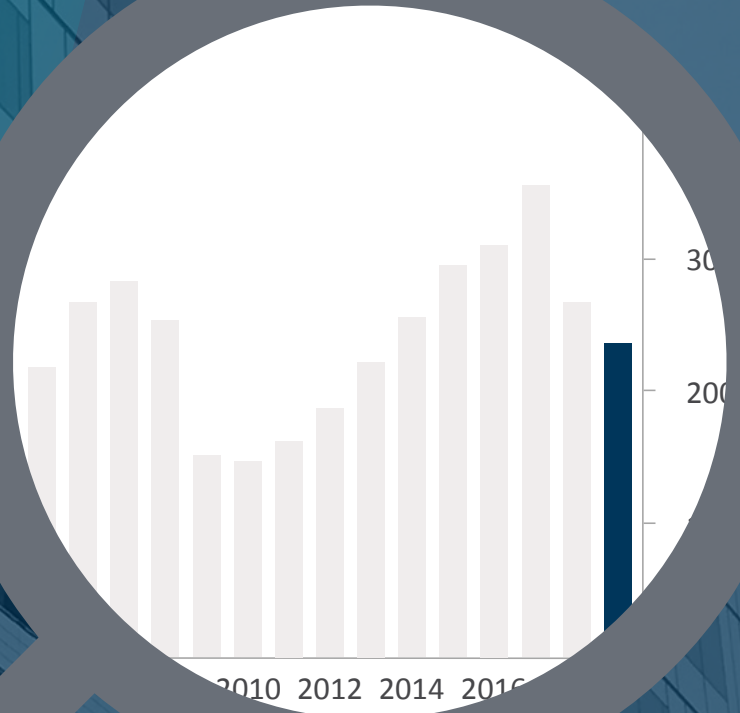
Example 2

Focusing attention on a data series in a 100% stacked chart; and adding grand totals



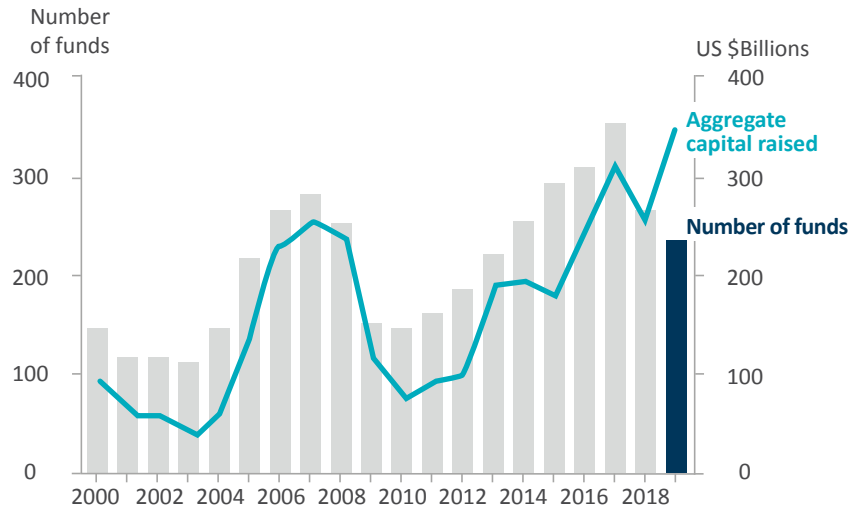
Use the colour midnight (or ocean) to highlight the important category. Colour the other segments Gray 3. The grand total series is gray 2, and its labels Text black. This example combines the 100% stacked bar with a total volume chart.

Chart Examples



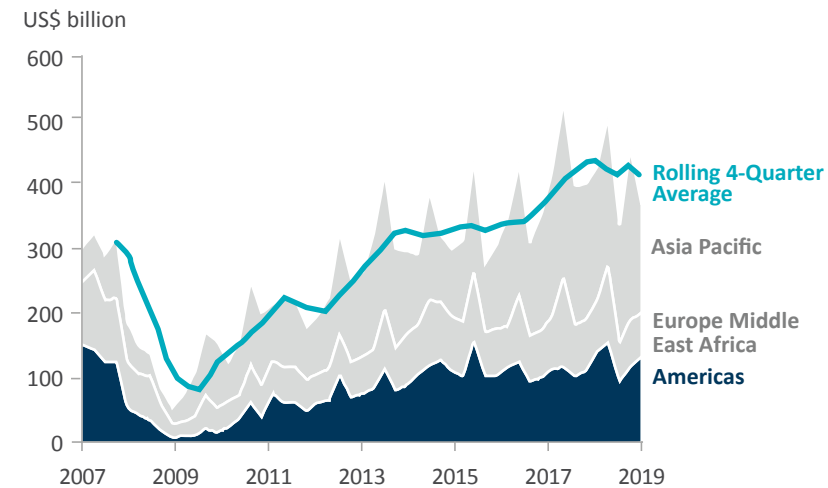
Stacked Charts

These next few pages present some additional formatting examples



Source: Prequin

Dual axis chart, featuring bar and line chart styles. The de-emphasized columns are shaded in Gray 2 — allowing the ocean coloured line to standout



Source: Real Capital Analytics

A combination chart featuring area and line chart styles. The de-emphasized areas are shaded in Gray 2.

FIGURE #

Rolling One Year Asset Class Net Returns and Fees (basis points)

Sub-titles can add explanatory and interpretive power to a graph.
Use them to answer questions a viewer might have or to highlight specific data points.

1 Year Net Return | 5 Year Net Return | 1 Year Fees

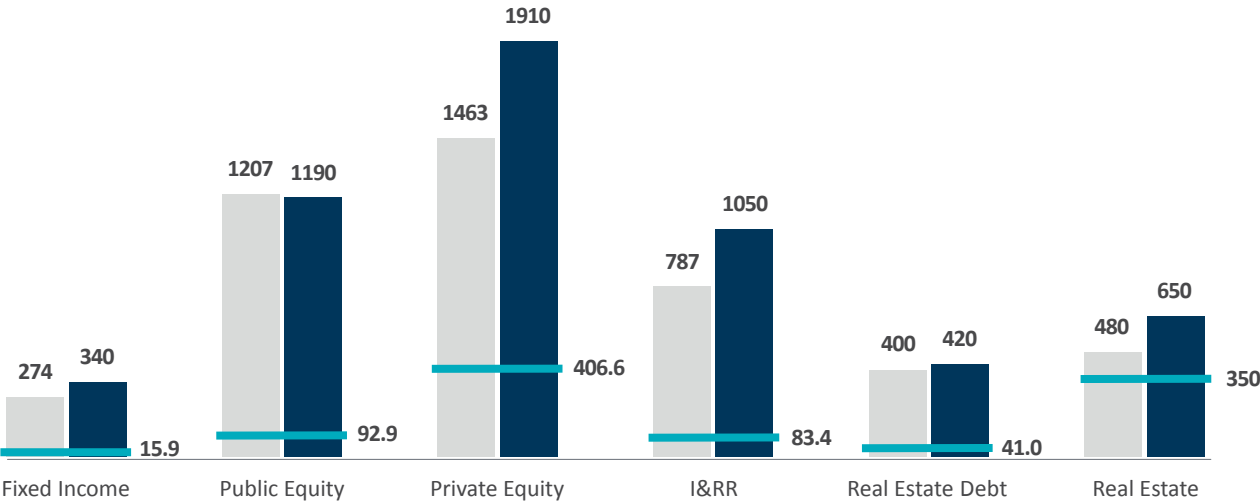


FIGURE #
Public Market Trailing One Year Net Returns and Fees (bps)

1 Year Net Return | 5 Year Net Return | 1 Year Fees

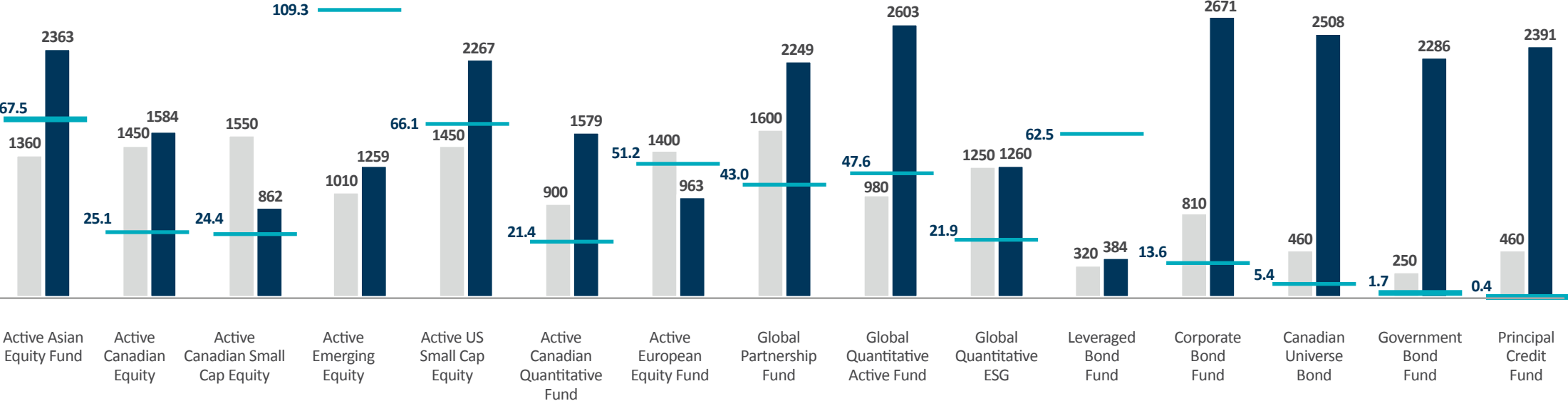


FIGURE #

WTI Spot Price

Sub-titles can add explanatory power to a chart; they can highlight key takeaways.

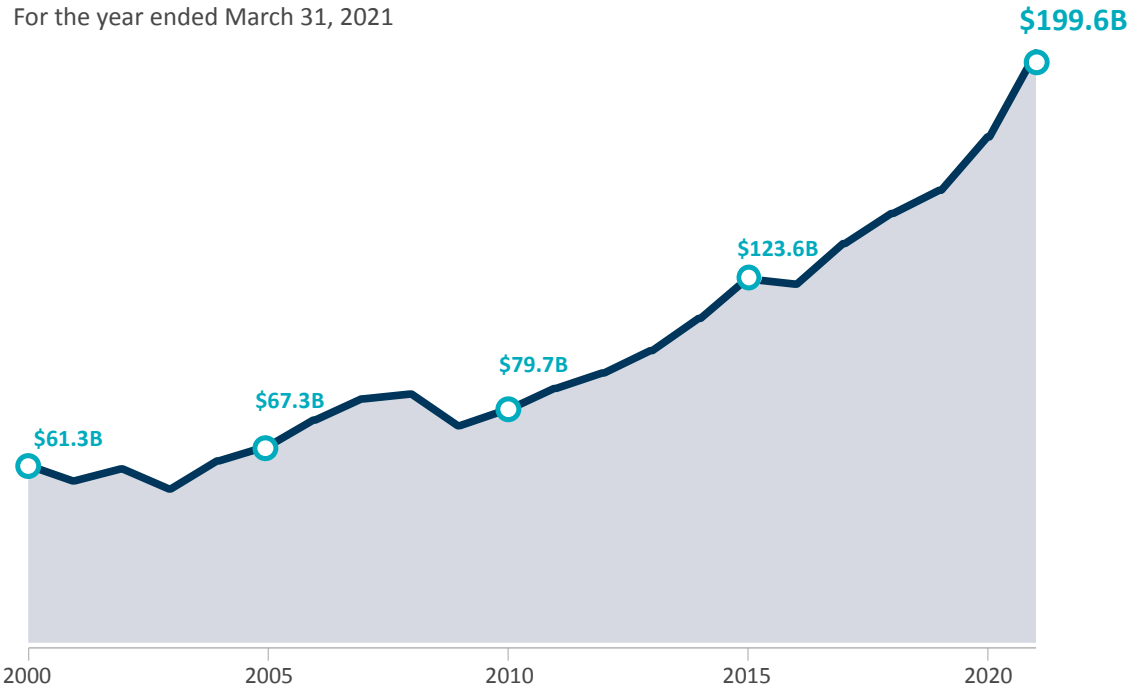


This line chart features shading in Gray 1, to draw focus to that period in time.

FIGURE #

Total assets under management (\$Billion)

For the year ended March 31, 2021



This line/area chart presents the significant story of BCI's AUM over 21 years. Subtle shading of midnight 10% helps to communicate the growth, without dominating the chart. Instead, ocean markers are used to highlight the AUM in five year increments.

FIGURE #

Short catchy title with first word capitalized

Sub-titles can add explanatory and interpretive power to a graph. Use them to answer questions a viewer might have or to highlight specific data points.

BCI Canadian Pool Funds | Benchmark | S&P/TSX Capped Composite Index

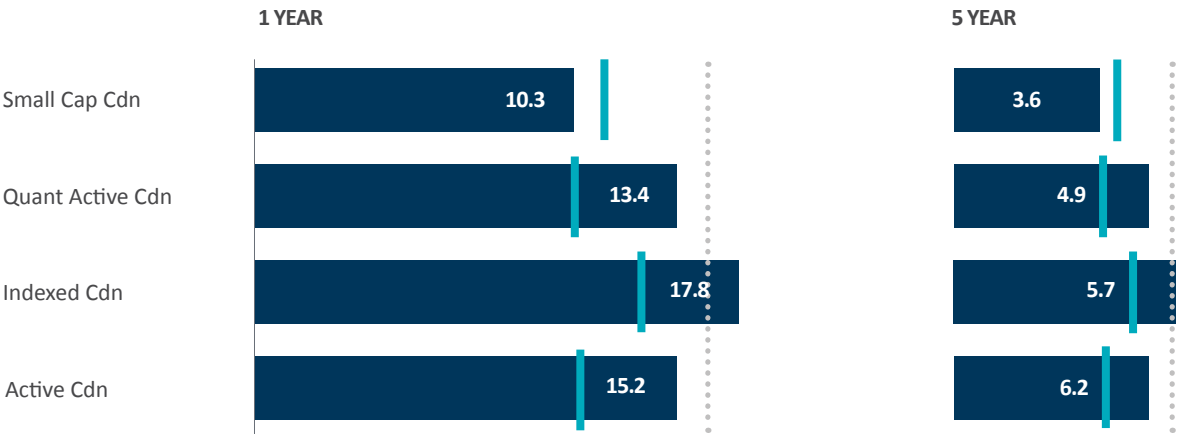
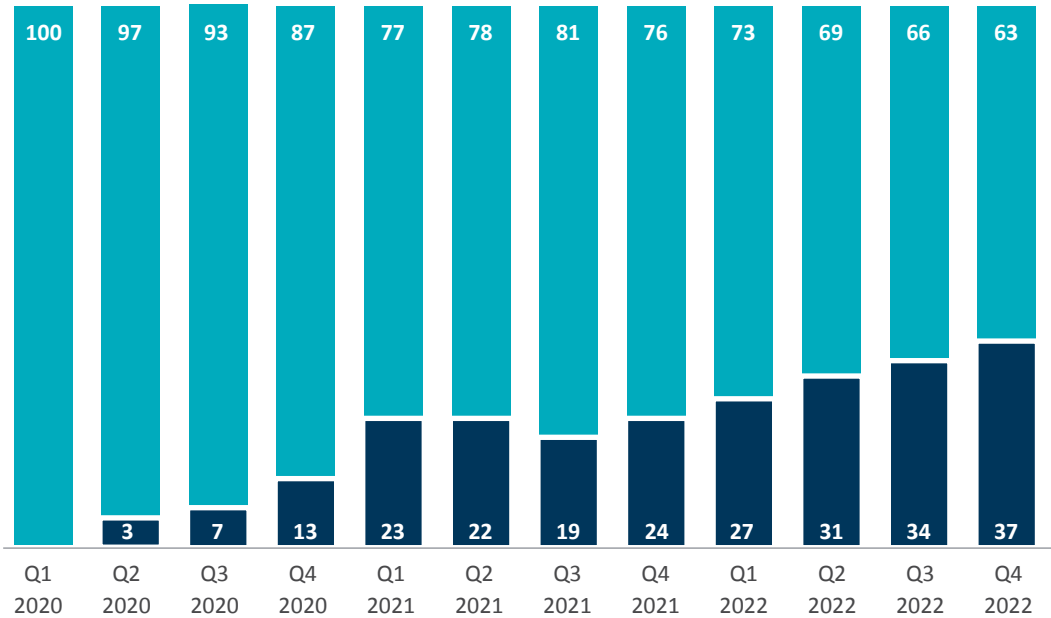


FIGURE #

Short catchy title with first word capitalized (%)

Sub-titles can add explanatory and interpretive power to a graph. Use them to answer questions a viewer might have or to highlight specific data points.

Co-investment/Direct deals | Funds



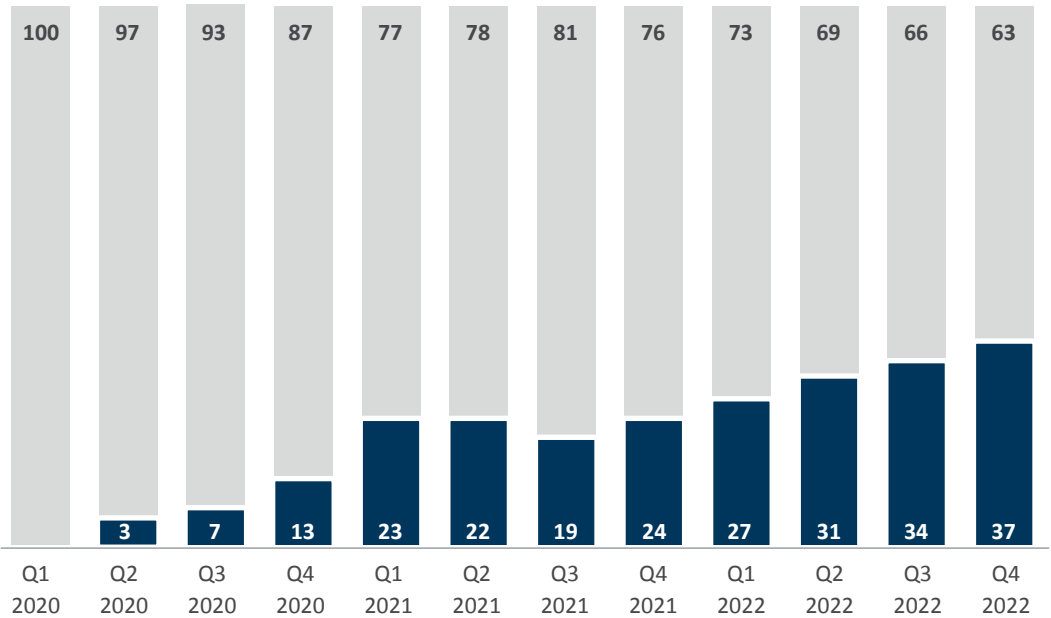
This chart’s use of colour gives both “Co-investments” and “Funds” equal prominence.

FIGURE #

Short catchy title with first word capitalized (%)

Sub-titles can add explanatory and interpretive power to a graph. Use them to answer questions a viewer might have or to highlight specific data points.

Co-investment/Direct deals | Funds



This chart features the “Co-investments” category, giving it more visual prominence than the funds category.



Tables

PUBLIC MARKETS

Fixed Income

Public Equities

Other Strategies

PRIVATE MARKETS

Infrastructure & Renewable Resources

Private Equity

QUADREAL PROPERTY GROUP

Real Estate

Mortgages

Tables

A well-designed table provides clarity on the information being presented and helps our audience make insights or take action.

GUIDELINES

Table width: A table should span 1, 2, or 3 columns on a page, so that they align with the other content.

Table borders: Do not put a border around a table.

Text and Number Alignment: Left-align text in a column; right-align numbers on the decimal point.

Table body text: Either Sentence case, or Title Case, avoid uppercase.

- Negative numbers displayed in () parentheses;
- Null values displayed as ‘en-dash’ (–);
- Use “n/a” to show data is not applicable;
- Use short, descriptive row and column headers. Use abbreviations if the audience will understand what they mean, or footnote and spell out/explain

Table headings: Use uppercase or Title Case. Whichever you choose, be consistent throughout the report.

Font size: preference is Calibri 9pt. If possible keep table text font size consistent in all tables within a report.

TABLE STYLE #1

FIGURE #

Short catchy title with first word capitalized

Sub-titles can add explanatory power to a chart.

	\$ BILLION	%
PUBLIC MARKETS		
Fixed Income	57.1	33.4
Public Equities	55.7	32.5
Other Strategies	(9.7)	(5.7)
PRIVATE MARKETS		
Infrastructure & Renewable Resources	18.3	10.7
Private Equity	17.9	10.4
QUADREAL PROPERTY GROUP		
Real Estate	25.5	14.9
Mortgages	6.5	3.8
NET ASSETS UNDER MANAGEMENT	171.3	100.0

SUB-HEADINGS:
Calibri bold midnight
(#00365B), 9pt, UPPERCASE,
FILL: Gray 2 (#D9D9D9)
OUTLINE: none

BODY TEXT ROWS:
Calibri 9pt, Text Black
(#48484A), Sentence
case (or Title Case where
appropriate)

HEADINGS: Calibri bold
midnight (#00365B), 9pt,
UPPERCASE, or Title Case
(be consistent)

FILL: White/none (FFFFFF)

TABLE WIDTH: 1-, 2-, or
3-columns wide on a page

ROW HEIGHT: .3" tall

CELL MARGINS: .1" Top,
Bottom, Left, Right sides

ROW LINES: Gray 3
(#BFBFBF), 1 pt

BOTTOM/TOTAL ROW LINE:
Gray 3 (#BFBFBF), 2.25 pt

These three tables show variations on column design: the first option is the standard for table design; the second option uses Gray 1 to feature the data in the column; the third table introduces vertical lines. This style can increase the visual noise on a table, use in moderation.

STANDARD DESIGN:
Horizontal lines only; vertical lines can add visual noise

	\$ BILLION	%
PUBLIC MARKETS		
Fixed Income	57.1	33.4
Public Equities	55.7	32.5
Other Strategies	(9.7)	(5.7)
PRIVATE MARKETS		
Infrastructure & Renewable Resources	18.3	10.7
Private Equity	17.9	10.4
QUADREAL PROPERTY GROUP		
Real Estate	25.5	14.9
Mortgages	6.5	3.8
NET ASSETS UNDER MANAGEMENT	171.3	100.0

FEATURING A COLUMN OF DATA:
Fill is Gray 1 (#)

	\$ BILLION	%
PUBLIC MARKETS		
Fixed Income	57.1	33.4
Public Equities	55.7	32.5
Other Strategies	(9.7)	(5.7)
PRIVATE MARKETS		
Infrastructure & Renewable Resources	18.3	10.7
Private Equity	17.9	10.4
QUADREAL PROPERTY GROUP		
Real Estate	25.5	14.9
Mortgages	6.5	3.8
NET ASSETS UNDER MANAGEMENT	171.3	100.0

VERTICAL LINES:
(optional) vertical lines can add visual noise to the table

	\$ BILLION	%
PUBLIC MARKETS		
Fixed Income	57.1	33.4
Public Equities	55.7	32.5
Other Strategies	(9.7)	(5.7)
PRIVATE MARKETS		
Infrastructure & Renewable Resources	18.3	10.7
Private Equity	17.9	10.4
QUADREAL PROPERTY GROUP		
Real Estate	25.5	14.9
Mortgages	6.5	3.8
NET ASSETS UNDER MANAGEMENT	171.3	100.0

This table combines bar chart data — “spark bars”.
Positive value bars are 10% midnight, while negative
bars are 20% orange

#E5EBEE – 10% midnight

#F8E0D5 – 20% orange

ASSET CLASS	MARKET VALUE (\$)	FUND ASSET WEIGHT (%)	CURRENT POLICY TARGET		POLICY RANGE
			ABSOLUTE (%)	RELATIVE (%)	MIN-MAX (%)
Short-Term	1,073,713,233	2.7	2.0	0.7	0 - 10
Government Bonds	8,974,891,756	22.2	21.0	1.2	5 - 35
Corporate Bonds	2,067,602,430	5.1	5.0	0.1	0 - 10
Private Debt	1,353,908,408	3.4	4.0	-0.6	0 - 10
Mortgages	1,470,086,054	3.6	4.0	-0.4	0 - 10

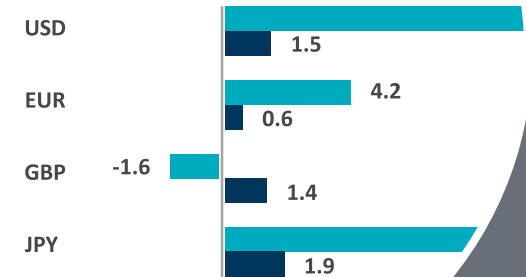
POSITIVE VALUES:
10% midnight or #E5EBEE

NEGATIVE VALUES:
20% orange or #F8E0D5

Sample Layouts

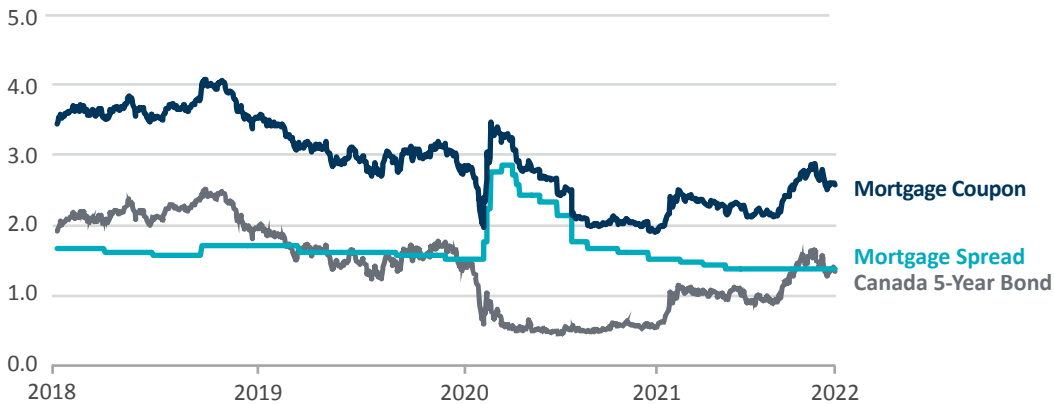
Currency and

Changes in FX (%)



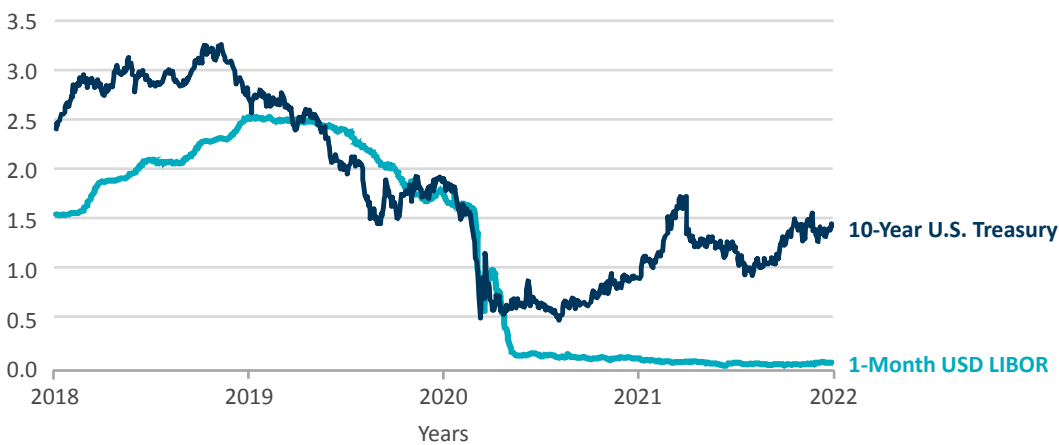
Positive indicates strengthening

Canadian mortgage spreads remain low (%)



Source: Bloomberg, CMLS

10-Year U.S. Treasury vs. LIBOR (%)



Source: Bloomberg, U.S. Treasury Department

Mortgages

REVIEW

- Velestia vendips untion elique modisti onsedit re vel ius. Uda coressum adiciistibus ex et et, ari dolore iligeni squatem porias receratur sitate non conestrum voluptur sit magnam, nusam fugia quis doluptur, sae.
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OUTLOOK

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POSITIONING

- Velestia vendips untion elique modisti onsedit re vel ius. Uda coressum adiciistibus ex et et, ari dolore iligeni squatem porias receratur sitate non conestrum voluptur sit magnam, nusam fugia quis doluptur, sae.
- Volupieni ullorep eruptaspel mo et odis escimenda aliamenda velluptati suna delessi dolenti sit audant ullataspel molorem inctis assera quam, acesequ iduntot aspistiure maio. Nam, si tem inctur?

Fiduciary Checklist

✓	Funding Ratio and Risk 20 Words aequē altera graeci ei, pri velit numquam. Quot possim feugiat ex his, cibo oblique antiopam sed, facilisis.
✓	Liquidity 20 Words aequē altera graeci ei, pri velit numquam. Quot possim feugiat ex his, cibo oblique antiopam sed, facilisis.
✓	Fees 20 Words aequē altera graeci ei, pri velit numquam. Quot possim feugiat ex his, cibo oblique antiopam sed, facilisis.
▲	ESG 20 Words aequē altera graeci ei, pri velit numquam. Quot possim feugiat ex his, cibo oblique antiopam sed, facilisis.
✓	Asset Allocation 20 Words aequē altera graeci ei, pri velit numquam. Quot possim feugiat ex his, cibo oblique antiopam sed, facilisis.

KEY TAKE-AWAYS

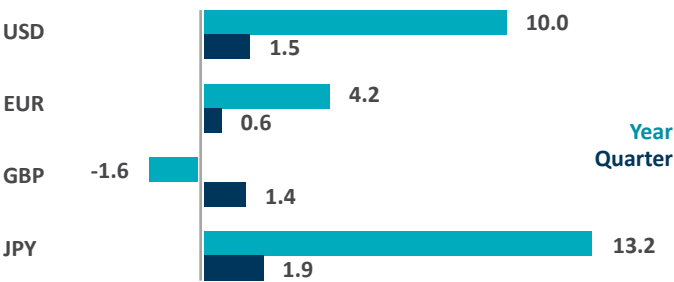
- 175 Words Performancesuscipiantur et. Vis ut erat reque iisque, pro eros prima delicatissimi ea, cu disputationi. At eum nostrum nominati. Cum ei timeam eleifend. Has ei vivendo delectus concludaturque. Usu solum iuvaret ea. Pro luptatum periculis cu, ius at error causae salutandi.
- Mea falli equidem dignissim ex. Recteque rationibus qui an, an usu nihil tation detraxit. Est aequē altera graeci ei, pri velit numquam ex. Quot possim feugiat ex his, cibo oblique antiopam sed id, est ne libris facilisis. Vix elaboraret percipitur et, ad vix fugit vivendo oporteat, ei sea viderer feugiat recusabo.
- Eum id voluptaria omittantur. Et equidem persecuti expetendis his, has decore tritani te. Atomorum tractatos ei eam, an est postea veritus. Commune percipitur vix ex, has alia saepe laboramus ut.
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3-column layout featuring risk symbol in a table

Currency and Fixed Income Dashboard

▲ Changes in FX (%)

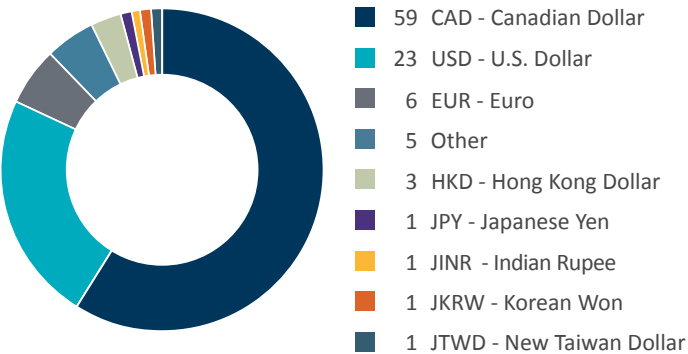


Positive indicates strengthening

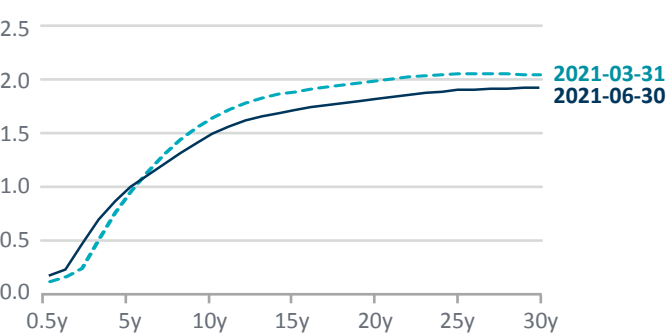
STRENGTHENING CAD

20 Words Hicipsant quates dus ati autem etur? Ximi, nos sim volorepudi vit lab int molupta con conetur Voluptae perchilla

Total Fund currency exposure (%)



✓ Canada Yield Curve (%)

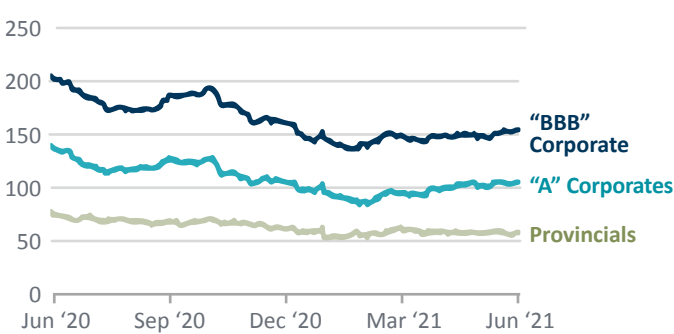


DECREASING LONG TERM YIELD

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✓ Corporate Credit Spreads (bps)



TIGHTENING CREDIT SPREADS

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Total Plan Contribution

PERFORMANCE SUMMARY

- 1

170 words

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Illustrative Point 3. Fusce aliquet pede non pede. Suspendisse dapibus lorem pellentesque magna. Integer nulla. Donec blandit feugiat ligula. Donec hendrerit, felis et imperdiet euismod, purus ipsum pretium metus, in lacinia nulla nisl eget sapien. Donec ut est in lectus consequat consequat. Etiam eget dui. Aliquam erat volutpat. Sed at lorem in nunc porta tristique. Quisque aliquam tempor.

